An extremal even unimodular lattice in dimension 112

Gluing with $\mathrm{SL}_2(29)$ symmetry and a computer-assisted proof

roiban1344
github.com/roiban1344
Draft, version v2026-11-02

Abstract

We construct an explicit positive-definite even unimodular lattice L_{112} of dimension 112 and minimum norm 10, attaining the extremal bound. The construction glues two copies of a 56-dimensional extremal lattice of Nebe using an endomorphism whose square is -5 and which commutes with $\mathrm{SL}_2(29)$. Extremality is reduced to finitely many coset bounds and proved by checking orbit completeness and either canonical forms or integer conditions derived from the midpoint lemma. We also determine the full automorphism group, of order 1 169 280; this part uses the classification of finite simple groups. Defining matrices, verification data and code are supplied, and the deduction from explicit computational hypotheses is formalized in Lean.

1. Introduction

1.1 Extremal even unimodular lattices

A positive-definite lattice of rank n is a discrete subgroup $\Lambda = \mathbb{Z}b_1 \oplus \ldots \oplus \mathbb{Z}b_n$ spanning Euclidean space $\mathbb{R}^n$. We write $\langle x, y\rangle$ for the inner product and call $\langle x, x\rangle$ the norm of x. The lattice is even if every norm is an even integer. An integral lattice is unimodular if its Gram matrix has determinant 1, or equivalently if it equals its dual. Even unimodular lattices exist only in dimensions divisible by 8, and an example exists in every such positive dimension [SPLAG, Ch. 7].

The theta series $\theta_{\Lambda(q)} = \sum_{x\in\Lambda} q^{\langle x,x\rangle/2}$ of an even unimodular lattice is a modular form of weight $\frac{n}{2}$. This gives the bound of Mallows–Odlyzko–Sloane [MOS75]:

$$\min\Lambda \le \mu(n) := 2\left\lfloor \frac{n}{24}\right\rfloor + 2. \tag{1.1}$$

Here $\min\Lambda$ is the smallest norm of a nonzero vector. An even unimodular lattice is extremal when equality holds in (1.1). Unless another class of lattices is explicitly specified, "extremal lattice" in this paper means extremal even unimodular lattice. We write $\mu(n)$ for this bound.

The bound is 6 in dimension 56 and 10 in dimension 112. Thus the construction below must exclude nonzero vectors of norms 2, 4, 6 and 8 in dimension 112. Examples in lower dimensions include E_8 and the Leech lattice. Dimension 88 was open until 2026: Sakamoto's preprint [Sak26] constructs two examples, and four further ones, ideal lattices over $\mathbb{Q}(\zeta_{115})$ from a preprint of Gauvin [Gau26], are now listed in the Catalogue of Lattices [NS]. Further background and comparison tables are collected in Appendix G; the proof of the result below does not depend on those comparisons.

1.2 Construction and main result

Let L_{56} denote the 56-dimensional lattice whose Gram matrix is the file `L56_gram.txt` listed in Appendix D. (The defining matrices are not printed in this paper; they are distributed as data files identified by their SHA-256 hashes. Appendix D also records a specification of L_{56} and τ in terms of Nebe's construction that does not depend on the files.) The lattice L_{56} is $\mathrm{SL}_2(29)$-equivariantly isometric to Nebe's lattice $L_{56,2}\big(\widetilde{M}\big)$ [Neb98, Thm. 2.4]; the identification is established in §7.2. The two matrices in the file `sl2_29_generators.txt` are isometries of L_{56} and generate a group S of order 24 360 (F4), isomorphic to $\mathrm{SL}_2(29)$ [C2]. Let τ be the endomorphism given by the file `tau.txt`. It commutes with S and satisfies

$$\tau^2 = -5\ \mathrm{id}, \quad \langle x\tau, y\tau\rangle = 5\langle x, y\rangle \quad (x, y \in L_{56}).$$

Matrices act on row vectors on the right. These identities and the defining matrices are checked by [C2]. Here and below, F0–F5 refer to the computational facts listed in §4.3, and bracketed labels such as [C2] to the rows of Table D2 in Appendix D, which name the code for each computation. For this choice of τ, set

$$L_{112} = L_\tau := \left\{ \frac{(x, x\tau + 2y)}{\sqrt{2}} : x, y \in L_{56} \right\} \subset \frac{1}{\sqrt{2}}(L_{56} \perp L_{56}). \tag{1.2}$$

> **Theorem A.** The lattice L_{112} defined by (1.2) from the data files `L56_gram.txt` and `tau.txt` of Appendix D is an even unimodular lattice of dimension 112 with minimum norm 10. In particular, it is extremal.

The diagonal action of S preserves L_{112}. The proof of Theorem A is outlined in §1.3 and completed in §4.3 using the finite computations in §6. The construction and the verification are given separately, so the verification does not require repeating the search that led to τ.

The choice of τ is not unique. The centralizer C of S in $\mathrm{Aut}(L_{56})$ has order 12, and exactly 24 endomorphisms of L_{56} commute with S and satisfy $\tau^2 = -5$ [C2]. Only the reduction of τ modulo $2L_{56}$ enters the glued lattice. Proposition 3.3 classifies these reductions into three classes of gluing maps; the 24 endomorphisms reduce into two of them, called B and C below, 12 into each [C2], and endomorphisms in the same class give isometric glued lattices. The supplied τ lies in class C, which gives L_{112}. Class B gives an even unimodular lattice with minimum norm 8 and exactly 292 320 vectors of norm 8 [C4]. This second lattice serves as a positive control: the verification programs must detect its short vectors. The classification of the gluings is completed in §3.3.

Two further conclusions are treated separately from the extremality proof. The extremal theta series determines that L_{112} has 10 888 335 360 vectors of norm 10 and 247 006 775 232 000 vectors of norm 12 (Appendix B). Its full automorphism group has order 1 169 280 (Theorem B, §8 and Appendix F); the proof uses the classification of finite simple groups. Theorem A does not use the determination of this full group.

1.3 Outline of the proof and the verification methods

We first verify that the lattice in (1.2) is even unimodular (Lemma 3.2). The main remaining task is to prove that it has no nonzero vector of norm below 10. The minimum norm of L_{56} is 6, as reported in [Neb98]; we do not rely on that value but check it directly from the data file by two rigorous enumerations, one using interval arithmetic and one using an integer quadratic form (§6.1).

Consequently, a nonzero vector $\frac{(x,v)}{\sqrt{2}}$ of norm below 10 in L_{112} would have both components in L_{56} and would satisfy

$$v \equiv x\tau \pmod{2L_{56}}, \quad (\langle x,x\rangle, \langle v,v\rangle) \in \{(6,6),(6,10),(10,6),(8,8)\}.$$

Lemma 4.2 shows that this list is exhaustive and that neither component is zero. Since $\tau^2 \equiv 1 \pmod 2$, interchanging the two components reduces the type $(10,6)$ to $(6,10)$ (Appendix C.4). It therefore suffices to establish the following two bounds:

$$\min(r\tau + 2L_{56}) \geq 14 \quad \text{for all } r \in L_{56}(6),$$

$$\min(x\tau + 2L_{56}) \geq 12 \quad \text{for all } x \in L_{56}(8),$$

where $L_{56}(m)$ denotes the vectors of norm m in L_{56} and the minimum of a coset is taken over all its vectors. This is Corollary 4.3.

Because the group S preserves norms and commutes with τ, each bound need only be checked on a collection of orbit representatives. For norm 6 there are 656 orbits. For norm 8 the supplied collections give 1 527 984 test vectors of L_{56}, with repetitions, covering 1 517 130 distinct orbits. These are vectors of the 56-dimensional lattice, not norm-8 vectors of L_{112}. The completeness check verifies the norms, disjointness and orbit sizes, and compares the total sizes with the shell sizes determined by the theta series (§§6.2–6.4).

The lower bounds are then verified as follows. For each norm-6 representative, we supply a vector in its prescribed coset whose norm and inner products with all minimal vectors of L_{56} imply the bound 14. The implication follows from the midpoint lemma and Corollary 5.2; the supplied vector and its conditions are checked in §6.6. For the norm-8 test vectors, there are two alternatives. One uses canonical forms to test whether the prescribed coset contains a norm-8 vector (§6.5). The other checks coset representatives whose norm and inner products imply the bound 12 (Corollary 5.3 and §6.7). Either alternative, together with the common norm-6 bounds and completeness checks, proves the required absence of short vectors. The upper bound (1.1) then gives $\min L_{112} = 10$.

Thus the two even-coset tests are alternative ways of completing the same proof, and their shared inputs are part of the verification. Separately written implementations check these inputs and computations. Spherical-design identities, changes of frame and tests on the second gluing provide further consistency checks. Floating-point closest-vector searches and random samples are also reported, but are not substituted for the rigorous checks in the proof. Supplement §S4 describes a further route through support modules. It replaces the reduction of §4 by Coulangeon's inequality and an enumeration of support modules, but it is not independent of the proof, since its last step uses the same computations F5a and F5b (together with a further finite check, Supplement [C32]). The Lean formalization checks the deduction from the stated computational hypotheses (§7.1).

Scale, and what the proof rests on. Enumerating the vectors of norm at most 8 of L_{112} directly would visit about 10^{25} enumeration nodes (Supplement §S3.2). The reduction above replaces this by an enumeration of 8.5×10^9 nodes in dimension 56 and by tests of 656 and 1 527 984 cosets; checking all computational facts from the distributed data takes about 1.2 CPU hours (Appendix D). The proof of Theorem A rests on (i) the hand proofs of Lemmas 3.2, 4.2 and 5.1, Corollaries 4.3, 5.2 and 5.3 and Proposition 6.3, the deduction from the computational facts being also checked in Lean (§7.1); (ii) the bound (1.1) and the extremal theta series in dimension 56 (§2.3); (iii) the correctness of the verification programs for F0–F5, which make every decision in integer arithmetic or in outward-rounded interval arithmetic (§6), and of the Python, NumPy and C compiler that run them; and (iv) the

identity of the four data files, fixed by their SHA-256 hashes (Appendix D). The other distributed data need not be trusted: the programs check again every property of them that the proof uses. Separately written implementations reproduce the computations (§6, Supplement §S3.3); since both were written by sessions of the same AI system and share the data files, they guard against independent programming errors but not against a shared misreading of the specification (see "Use of AI tools").

1.4 Relation to previous constructions and the choice of dimension

The construction uses established ingredients. The base lattice is one of Nebe's 56-dimensional lattices [Neb98, Thm. 2.4]. Gluing several copies of a lattice along a hyperbolic decomposition of L/pL goes back to Quebbemann [Que84]; with polarizations of $L/2L$ coming from Hermitian structures, this construction gives the extremal lattices of dimension 80 [BN98] and Γ_{72} [Neb12, Def. 2.6, Rem. 2.7], following Griess's analysis of rank-72 lattices built from the Leech lattice [Gri10] (see also [Gri99]). Our gluing (Definition 3.1) is of a different kind: it uses the graph of an isometry of $L/2L$ between two copies of the same lattice. The Hermitian tensor-product framework [Cou00, CN13] underlies extremal lattices in dimensions 48 (P_{48n} [Neb98]; [CN13, §4]), 72 [Neb12, CN13], 80 [BN98; Cou00, §4.2] and 88 [Sak26]. Proposition 3.4 identifies (1.2) with a Hermitian tensor product $H \otimes P$ over $\mathbb{Z}\left[\sqrt{-5}\right]$ with a unimodular Hermitian lattice H of rank 2, as for P_{48n} in [CN13, §4]. Unlike the base rings used there, $\mathbb{Z}\left[\sqrt{-5}\right]$ has class number 2, so $\mathcal{O}$-submodules of P need not be free; Supplement §S4 treats them through pseudo-bases. The result of this paper is the extremality of the lattice obtained with the supplied endomorphism, together with the explicit data and verification needed to establish it.

The closest earlier construction is Nebe's own: [Neb98, §5] tensors two lattices $L_{2(p-1),2}$ over a quaternion subring of their endomorphism rings and obtains the extremal lattice L_{48} for $(p, q) = (5, 13)$ [Neb98, Thm. 5.3] and a 64-dimensional lattice, proved extremal in a footnote, for $(5, 17)$. The pair $(5, 29)$ would give dimension $8 \cdot 56/4 = 112$, but it does not appear in [Neb98]. Indeed, over the maximal orders of $\mathcal{Q}_{\infty,2}$ and $\mathcal{Q}_{\infty,3}$ (for every choice of the two embeddings) the tensor product admits no unimodular invariant form, even after multiplying it by a fractional ideal of $\mathbb{Q}\left(\sqrt{5}, \sqrt{29}\right)$. The obstruction is this: if Λ is the tensor product and $\mathfrak{a}$ the ideal with $\Lambda^{\#} = \Lambda\mathfrak{a}$, such a twist can be unimodular only if the narrow ideal class of $\mathfrak{a}$ is a square. For $(5, 29)$ the class group and the narrow class group of $\mathbb{Q}\left(\sqrt{5}, \sqrt{29}\right)$ have order 2, so a square class is trivial, and $\mathfrak{a}$ is not principal; the corresponding fields for $(5, 13)$ and $(5, 17)$ have trivial class groups, also in the narrow sense [C33]. Our L_{112} arises by a different route: gluing two copies of the same $L_{56,2}\left(\widetilde{M}\right)$ along $L_{56}/2L_{56}$. Its Gram matrix (3.1) has the same shape as Nebe's "easy construction" of L_{48} from two copies of a 3-modular lattice and an endomorphism $1 + i$ of reduced norm 2 [Neb98, p. 497].

Reductions of extremality to finite computations on cosets modulo $2L$ are classical. The midpoint lemma (Lemma 5.1) is the standard argument bounding the short vectors in a class of $L/2L$, used for the Leech lattice in [Neb12, Lemma 3.1]; Nebe's proof that Γ_{72} has minimum 8 reduces to computations with the minimal vectors of the Leech lattice [Neb12], and Nebe and Parker showed by computer search that Γ_{72} is the only extremal lattice obtainable from Griess's construction [NP14]. Kominers [Kom09] showed that extremal even unimodular lattices of dimension 56, 72 or 96 are generated by their minimal vectors. What is new here on the computational side is the scale of the reduction (656 odd and 1 527 984 even classes), the certificate form of Corollaries 5.2 and 5.3, which turns each coset bound into a few integer checks, and the canonical-form test on $L_{56}/2L_{56}$, each checked again by separately written programs from the distributed data. We have no conceptual explanation of why class C gives minimum 10 while class B gives minimum 8 (both lift to elements τ with $\tau^2 = -5$ and differ only modulo $2M'$).

Supplement §S4 gives a partial one in terms of the Hermitian structure: a vector of norm 8 in L_τ gives a τ-stable $\mathcal{O}$-submodule of rank 2 of P_τ with determinant at most 4 (Coulangeon's inequality), and the norm-8 vectors of gluing B come from two types of such submodules, one of them attaining Coulangeon's bound (Supplement Table S9).

The choice of dimension 112 was motivated by comparison with dimensions 96 and 104, where the extremal bound is also 10. The Siegel weighted mean of the number of nonzero vectors of norm below 10 is approximately 15 732 086, 1 046 569 and 51 014 in these three dimensions, respectively [C1]. The smaller mean in dimension 112 suggested a useful direction for the search. These are averages over a genus, not estimates of the time needed to prove extremality. The available $\mathrm{SL}_2(29)$ action on L_{56} also supplies the symmetry used to reduce the verification to orbit representatives. Supplement §S1 gives the numerical comparisons and the search motivation.

1.5 Organization of the paper

Sections 2–6 prove Theorem A: notation, construction, reduction to four types, coset bounds and finite computations, respectively. Section 7 describes the scope of the Lean formalization and identifies the base lattice with Nebe's lattice. Section 8 determines the full automorphism group. Appendices A–G collect terminology, theta series, auxiliary proofs, data and code, pseudocode, the group-theoretic proof and known examples.

A separate supplement contains the Siegel means and search motivation (Supplement §§S1–S2), verification routes, running times and additional tests (Supplement §S3), the support-module argument (Supplement §S4), a comparison of constructions considered in the search (Supplement §S5), and the classification of the even unimodular lattices of dimension 112 on which $\mathrm{SL}_2(29)$ acts as on L_{112} (Supplement §S6, Theorem S6.1). Its sections and tables carry the prefix S.

2. Preliminaries and terminology

We fix conventions for lattices, cosets, group actions and Hermitian forms. Norm always means squared length.

2.1 Conventions and terminology

We use L_{56} for the fixed lattice of §1; the letter L in general definitions and lemmas denotes an arbitrary lattice satisfying the stated assumptions. An isometry between lattices is an additive-group isomorphism preserving the inner product; it need not preserve orientation.

Lattices are given by Gram matrices $G \in \mathbb{Z}^{n\times n}$ with respect to a basis; vectors are integer **row** vectors x, with $\langle x, y\rangle = xGy^{\mathsf{T}}$. A matrix g acts by $x \mapsto xg$; it is an isometry if $gGg^{\mathsf{T}} = G$. "Norm" always means $\langle x, x\rangle$, never its square root. For a lattice L, the shell of norm m is $L(m) = \{x \in L : \langle x, x\rangle = m\}$.

Table 1. Terms distinguished in this paper.

term	symbol	meaning
minimum (of a lattice)	$\min \Lambda$	smallest norm of a nonzero vector of a specific lattice Λ
extremal bound	$\mu(n)$	the upper bound $2\lfloor n/24 \rfloor + 2$ for the minimum in dimension n; $\mu(56) = 6$, $\mu(112) = 10$
inner-shell vector	—	nonzero vector of norm $< \mu(n)$; extremal $\iff$ none exists
coset minimum	$\min(c)$	smallest norm of a vector in a coset c modulo $2L$ (§2.2)
centralizer	C	elements of $\mathrm{Aut}(L)$ commuting with every element of $\mathrm{SL}_2(29)$ (§2.4); not the full unit group of an order

2.2 Cosets modulo $2L$

For an even unimodular L, the quotient $L/2L$ is an $\mathbb{F}_2$-vector space of dimension n; we call its elements $c = x + 2L$ **cosets**. Since $\langle x+2y, x+2y\rangle = \langle x,x\rangle + 4\langle x,y\rangle + 4\langle y,y\rangle$, the function $q(x+2L) = \langle x,x\rangle/2 \bmod 2$ is a well-defined quadratic form on $L/2L$. We call a coset *even* or *odd* according to q; all vectors in an even coset have norm $\equiv 0 \pmod 4$, and all vectors in an odd coset have norm $\equiv 2 \pmod 4$. A linear map f on $L/2L$ is **parity-preserving** if $q(cf) = q(c)$ for all cosets. In dimension 56 there are $2^{55} + 2^{27}$ even cosets (including 0) and $2^{55} - 2^{27}$ odd cosets (these are the numbers of zeros and of nonzeros of a nondegenerate quadratic form of plus type).

2.3 Extremal theta series and shells of L

If L is extremal of dimension 56, its theta series is

$$\theta_L = 1 + 15\,590\,400\,q^3 + 36\,957\,286\,800\,q^4 + 15\,284\,192\,071\,680\,q^5 + \cdots$$

[C1] (by [Ser73, Ch. VII], the space of modular forms of weight 28 is 3-dimensional, so "constant term 1, no q, no q^2" determines the form; Appendix B). Thus $|L(6)| = 15\,590\,400$, $|L(8)| = 36\,957\,286\,800$, $|L(10)| = 15\,284\,192\,071\,680$. Moreover each nonempty shell of an extremal lattice of dimension $24k+8$ is a spherical 7-design (Venkov [Ven84]; see also [BV01]); we use this only as a consistency test (§6.3).

2.4 Nebe's lattices with $\mathrm{SL}_2(29)$ and the centralizer

The group $\mathrm{SL}_2(29)$ has a faithful irreducible representation of degree 14 whose character field is $\mathbb{Q}(\sqrt{29})$ and which has Schur index 2 over $\mathbb{Q}$; as a rational representation (Nebe's Δ_2) it has degree 56, and its endomorphism algebra is the totally definite quaternion algebra $\mathcal{Q} = \mathcal{Q}_{\sqrt{29},\infty,\infty}$ over $\mathbb{Q}(\sqrt{29})$ ramified only at the two infinite places [Neb98, Lemma 2.2(i)]. Nebe constructs $\mathrm{SL}_2(29)$-invariant even unimodular lattices $L_{56,2}(M)$ and $L_{56,2}(\widetilde{M})$ attached to the two conjugacy classes of maximal orders $M, \widetilde{M}$ of $\mathcal{Q}$ [Neb98, Thm. 2.4] (we write $M, \widetilde{M}$ for Nebe's $\mathfrak{M}, \widetilde{\mathfrak{M}}$). Following [Neb98, p. 483], M is the class containing a maximal order of the rational quaternion algebra $\mathcal{Q}_{\infty,2}$ ramified at 2 and ∞ (the Hurwitz order), and $\widetilde{M}$ the class containing a maximal order of $\mathcal{Q}_{\infty,3}$; equivalently, the units of reduced norm 1 form a group of order 24 in M and of order 12 in $\widetilde{M}$ [C33]. Nebe lists the minimum of both lattices as 6 [Neb98, Table II]. For $L_{56,2}(M)$ the table also gives the 15 590 400 minimal vectors, which were enumerated from a basis reduced by Napias's LLL algorithm over Euclidean domains [Neb98, p. 483]; for $L_{56,2}(\widetilde{M})$ it gives the minimum only. The Catalogue of Lattices [NS] lists minimum 6 and kissing number 15 590 400 for both. To give a proof whose completeness is guaranteed including rounding errors and which depends neither on these values nor on the identification of §7.2, we verify the minimum of L_{56} directly from the distributed data (§6.1). The proof of Theorem A does not use the value from [Neb98].

For an $\mathrm{SL}_2(29)$-invariant lattice L we write $M' = \mathrm{End}_{\mathrm{SL}_2(29)}(L)$ for the ring of integer matrices commuting with $\mathrm{SL}_2(29)$ and preserving L ($\mathbb{Z}$-rank 8). Its elements that are isometries of L form the **centralizer** $C = C_{\mathrm{Aut}\,L}(\mathrm{SL}_2(29))$. The adjoint defined by the inner product agrees with the standard quaternion involution, so $uu^* = 1$ is equivalent to reduced norm 1. The centralizer C is the finite group of elements of reduced norm 1 of M'; it has order 24 for $L_{56,2}(M)$ and order 12 (6 up to sign) for the lattice L_{56} used here [C2].

2.5 Hermitian lattices and trace forms

Let $K = \mathbb{Q}(\sqrt{-5})$ with ring of integers $\mathcal{O} = \mathbb{Z}[\sqrt{-5}]$ (class number 2). If τ is an endomorphism of a lattice L with $\tau^2 = -5$ and $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$, then L becomes an $\mathcal{O}$-module P_τ via $\sqrt{-5}\cdot x := x\tau$, and

$$h(x,y) = \langle x, y\rangle + \frac{\langle x\tau, y\rangle}{\sqrt{-5}}$$

is a Hermitian form (linear in the first variable) with $\mathrm{Re}\, h = \langle\cdot,\cdot\rangle$ (Appendix C.1). For Hermitian $\mathcal{O}$-lattices A, B the tensor product $A \otimes_{\mathcal{O}} B$ carries the real form $\mathrm{Re}(h_A \otimes h_B)$; this is the Hermitian tensor product of [Cou00, CN13].

3. The construction

We glue two copies of a 56-dimensional lattice along a correspondence of cosets modulo $2L$ to obtain a 112-dimensional lattice (§3.1). Listing the gluings of two copies of one base lattice along maps commuting with $\mathrm{SL}_2(29)$ gives only five cases (§3.3, Table 2): two have inner-shell vectors forced by symmetry, two have minimum 8, and the remaining one is our L_{112}. Finally we rewrite L_{112} as a Hermitian tensor product (§3.4).

3.1 Gluing two copies modulo 2

Definition 3.1 (gluing). Let L be even unimodular of dimension n and f an invertible parity-preserving linear map on $L/2L$. The **gluing** of two copies of L along f is

$$L_f = \left\{ \frac{1}{\sqrt{2}}(x,v) : x, v \in L,\ v + 2L = (x + 2L)f \right\}, \qquad \left\langle \frac{(x,v)}{\sqrt{2}}, \frac{(x,v)}{\sqrt{2}} \right\rangle = \frac{1}{2}(\langle x, x\rangle + \langle v, v\rangle).$$

We call f the gluing map and L_f the glued lattice. Taking $L = L_{56}$, when f is the reduction of an integer matrix τ with $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$, we get the lattice L_τ of (1.2).

Lemma 3.2. Let L be even unimodular of dimension n and $\tau \in \mathrm{End}(L)$ with $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$. Then L_τ is an even unimodular lattice of dimension $2n$.

Proof. The map $L \oplus L \to L/2L$, $(x,v) \mapsto v - x\tau + 2L$ is a surjective homomorphism (already on $x = 0$). Its kernel L'_τ has index 2^n in $L \oplus L$, so $\det L'_\tau = 2^{2n}\det(L \perp L) = 2^{2n}$, and scaling by $1/\sqrt{2}$ multiplies the $2n$-dimensional Gram determinant by 2^{-2n}; hence $\det L_\tau = 1$. For evenness write $v = x\tau + 2y$:

$$\langle x, x\rangle + \langle v, v\rangle = \langle x, x\rangle + 5\langle x, x\rangle + 4\langle x\tau, y\rangle + 4\langle y, y\rangle = 6\langle x, x\rangle + 4(\langle x\tau, y\rangle + \langle y, y\rangle) \equiv 0 \pmod 4,$$

since $\langle x, x\rangle$ is even. So $\frac{1}{2}(\langle x, x\rangle + \langle v, v\rangle)$ is even. A lattice all of whose norms are even integers is integral by polarization. □

The basis $(e_i, e_i\tau)/\sqrt{2}$, $(0, 2e_i)/\sqrt{2}$ $(1 \le i \le 56)$ of L_τ has Gram matrix

$$\begin{pmatrix} 3G & \tau G \\ (\tau G)^T & 2G \end{pmatrix}, \tag{3.1}$$

which is the form in which L_{112} is distributed (Appendix D).

3.2 Forced collisions from the centralizer

If f agrees modulo 2 with an element $u \in C$, then $(x, xu)/\sqrt{2} \in L_f$ for all x, and a minimal vector x of L_{56} gives an inner-shell vector of norm $\frac{1}{2}(6+6) = 6$. We call this a **forced collision**: it is caused by symmetry, not by chance. Examples of such constructions are discussed in Supplement §S5.

3.3 Equivariant gluings of two copies of one lattice

In this section L is one of the two base lattices $L_{56,2}(M)$ and L_{56}, and $M' = \mathrm{End}_{\mathrm{SL}_2(29)}(L)$ (§2.4). To preserve the $\mathrm{SL}_2(29)$-symmetry, we glue two copies of the same L and require that f commute with $\mathrm{SL}_2(29)$ acting on $L/2L$; then the diagonal action of $\mathrm{SL}_2(29)$ preserves L_f. The ring of $\mathbb{F}_2$-linear maps on $L/2L$ commuting with $\mathrm{SL}_2(29)$ has $\mathbb{F}_2$-dimension 8 and equals the image of $M'/2M'$ [C3], so every commuting map is the reduction of an element of M'. The ring M' is a maximal order of $\mathcal{Q}$: it is an order containing the maximal order of Nebe's construction, namely M for $L_{56,2}(M)$ and $\widetilde{M}$ for L_{56} (by the identification of §7.2 [C23]). Since 2 is inert in $\mathbb{Q}(\sqrt{29})$ and $\mathcal{Q}$ is unramified at 2, $M'/2M' \cong M_2(\mathbb{F}_4)$, and the reduced norm modulo 2 becomes the determinant, with values in $R/2R \cong \mathbb{F}_4$, where R is the ring of integers of $\mathbb{Q}\big(\sqrt{29}\big)$.

Proposition 3.3. (a) An element of $M'/2M'$ is parity-preserving if and only if its reduced norm is 1. These elements are invertible and form the group $\mathrm{SL}_2(\mathbb{F}_4) \cong A_5$ of order 60. (b) $L_f \cong L_{ufu'}$ for $u, u' \in C$, and $L_f \cong L_{f^{-1}}$. (c) Modulo (b), the 60 maps fall into 2 classes, of sizes 12 and 48, for $L_{56,2}(M)$ and 3 classes, of sizes 6, 36 and 18, for our L_{56} (Table 2).

Proof. (a) Let $m \in M'$ and $c = \mathrm{nrd}(m) - 1 \in R$. The adjoint of right multiplication by m is right multiplication by $\bar{m}$ (§2.4), so $\langle xm, xm\rangle = \langle x, xm\bar{m}\rangle = \langle x, x\rangle + \langle x, xc\rangle$; both $\mathrm{nrd}(m) \bmod 2$ and parity preservation depend only on $m \bmod 2M'$. If m preserves parity, then $\langle x, xc\rangle \equiv 0 \pmod 4$ for all $x \in L$. As c is self-adjoint, polarizing gives $2\langle x, yc\rangle \equiv 0 \pmod 4$, so $Lc \subset 2L^{\#} = 2L$; hence $c/2 \in M' \cap \mathbb{Q}\big(\sqrt{29}\big) = R$. Conversely, if $c \in 2R$, write $c/2 = a + \bar{a} = \mathrm{trd}(a)$ with $a \in M'$ (the reduced trace of the maximal order M' maps onto R: its image is an ideal containing $2R$, and at 2 it is the trace of $M_2(R_2)$); then $\langle x, xc\rangle = 2\langle x, x(a+\bar{a})\rangle = 4\langle x, xa\rangle$. So parity preservation is $\mathrm{nrd}(m) \equiv 1 \pmod 2$, i.e. $\det = 1$ in $M_2(\mathbb{F}_4)$, and $|\mathrm{SL}_2(\mathbb{F}_4)| = 60$. (b) The map $(x, v) \mapsto (xu, vu')$ maps L_f isometrically onto $L_{u^{-1}fu'}$, and $(x, v) \mapsto (v, x)$ maps L_f onto $L_{f^{-1}}$. (c) The centralizer C consists of the units of M' of reduced norm 1 (§2.4), so its image $\bar{C}$ lies in $\mathrm{SL}_2(\mathbb{F}_4)$. The kernel of $C \to \bar{C}$ is $\{\pm 1\}$: for $u \in C$ other than ± 1, $\mathrm{trd}(u)$ has absolute value below 2 at both real places (u has reduced norm 1 and $\mathcal{Q}$ is definite); if $u \equiv 1 \pmod{2M'}$, then $\mathrm{trd}(u) \in 2R$, hence $\mathrm{trd}(u) = 0$ and $u^2 = -1$, which contradicts $u^2 \equiv 1 \pmod{4M'}$. So $|\bar{C}| = |C|/2$ is 12 for $L_{56,2}(M)$ and 6 for L_{56} (§2.4), and $\bar{C}$ is A_4 or S_3 (the subgroups of A_5 of these orders; A_5 has no element of order 6). By (b), the classes are the double cosets $\bar{C}g\bar{C}$, $g \in A_5$, each merged with its image under inversion. For $\bar{C} = A_4$, the stabilizer of a point in the action of A_5 on 5 points, the double cosets correspond to the 2 orbits of A_4 on the 5 points and have sizes 12 and 48. For $\bar{C} = S_3$, the stabilizer of a 3-subset T_0 in the action of A_5 on the 10 subsets of size 3, the orbits of S_3 are given by $|T \cap T_0| = 3, 2, 1$, of sizes 1, 6, 3, and the double cosets have sizes 6, 36, 18. Inversion preserves sizes, so it fixes every double coset. The enumeration [C3] confirms (a) and (c). □

Table 2. The five classes of gluing maps $f \in M'/2M'$ (Proposition 3.3) for the $\mathrm{SL}_2(29)$-equivariant gluings L_f of two copies of one lattice with $\mathrm{SL}_2(29)$. The rows are classes of gluing maps, not isometry classes of lattices: gluings A and B are isometric.

name	base	class of f (size)	$\min L_f$	inner-shell vectors
—	$L_{56,2}(M)$	contains an element of C (12)	6	forced collision
A	$L_{56,2}(M)$	does not (48)	8	292 320 of norm 8 [C4]
—	L_{56}	contains an element of C (6)	6	forced collision
B	L_{56}	does not (36)	8	292 320 of norm 8 [C4]
C	L_{56}	does not (18)	**10 (Theorem A)**	**none**

Both surviving classes B and C on L_{56} lift to elements $\tau \in M'$ with $\tau^2 = -5$. Of the 24 such elements, 12 reduce modulo 2 into class B and 12 into class C [C2]. Gluings A and B are isometric: an $\mathrm{SL}_2(29)$-equivariant isometry between them is supplied with the verification data and checked in integer arithmetic [C28]. Thus the two gluings of minimum 8 in Table 2 are the same lattice. Table 2 covers only gluings of two copies of the same base lattice along maps commuting with the diagonal action. Supplement §S6 treats all even unimodular lattices of dimension 112 on which $\mathrm{SL}_2(29)$ acts as on L_{112}, without assuming a gluing construction; this includes gluings of $L_{56,2}(M)$ with L_{56}, and gluings that intertwine the action on one copy with its twist by an outer automorphism (the twisted rational representation is isomorphic to the original one). Each such lattice is $L_{56,2}(M) \otimes_M \Lambda$ for a unimodular Hermitian lattice Λ of rank 2 over the maximal order M, and there are exactly eight such Λ up to isometry (Theorem S6.1) [C29, C30, C31]. Table 2 contains four of the resulting lattices. Of the other four, three contain a nonzero vector of norm at most 8. The minimum of the fourth, X_1 (a gluing of $L_{56,2}(M)$ with L_{56}), is not determined in this paper (Supplement §S6).

3.4 The Hermitian description

Proposition 3.4. Let H be the free Hermitian $\mathcal{O}$-lattice with basis e_1, e_2 and Gram matrix $\begin{pmatrix} 3 & \sqrt{-5} \\ -\sqrt{-5} & 2 \end{pmatrix}$ (so $\det H = 6 - 5 = 1$), and $P = P_\tau$ obtained from L_{56} as in §2.5. Then

$$\varphi : H \otimes_{\mathcal{O}} P \to L_\tau, \qquad e_1 \otimes x + e_2 \otimes y \mapsto \frac{1}{\sqrt{2}}(x,\, x\tau + 2y)$$

is an isometry for the real form $\mathrm{Re}(h_H \otimes h_P)$. (Appendix C.2)

This realizes L_{112} in the Hermitian tensor-product framework of [Cou00, CN13, Sak26]. Its full automorphism group is determined in §8.

4. Reduction to collisions

We show that an inner-shell vector of L_{112} corresponds to a pair (x, v), with $x \in L_{56}$ and v in the coset $x\tau + 2L_{56}$, whose norms are $(6,6)$, $(6,10)$, $(10,6)$ or $(8,8)$ (§4.1). The absence of such "collisions" can be expressed in terms of coset minima (Corollary 4.3), and the $\mathrm{SL}_2(29)$-symmetry reduces it to finitely many cases (§4.2, Table 3). We then state the computational facts F0–F5 on which the rest of the proof rests and derive Theorem A from them (§4.3); Figure 1 at the end of §4.3 shows how these facts and the lemmas combine.

4.1 Four types

From now on L_{56} is even unimodular of dimension 56 with $\min L_{56} = 6$ and τ satisfies $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$. The adjoint of τ is $\tau^* = 5\tau^{-1}$; since $\tau^2 = -5$, $\tau^* = -\tau$ (Appendix C.1).

Definition 4.1 (collision). For $a, b \in \{6, 8, 10\}$, a **collision of type (a, b)** is a pair (x, v) with $x \in L_{56}(a)$, $v \in L_{56}(b)$ and $v \equiv x\tau \pmod{2L_{56}}$. Equivalently, the coset $x\tau + 2L_{56}$ of some $x \in L_{56}(a)$ contains a vector of norm b.

Lemma 4.2. If $z = \frac{1}{\sqrt{2}}(x, v) \in L_\tau$ is an inner-shell vector ($z \neq 0$, $\langle z, z\rangle \leq 8$), then (x, v) is a collision of type $(6, 6)$, $(6, 10)$, $(10, 6)$ or $(8, 8)$.

Proof. If $x = 0$ then $v \in 2L_{56} \setminus \{0\}$, so $\langle v, v\rangle \geq 4 \cdot 6$ and $\langle z, z\rangle \geq 12$. If $v = 0$ then $x\tau \in 2L_{56}$; applying τ^* gives $5x \in 2L_{56}$, hence $x = 5x - 2 \cdot 2x \in 2L_{56}$ and again $\langle z, z\rangle \geq 12$. So $x, v \neq 0$ and $\langle x, x\rangle, \langle v, v\rangle \geq 6$. By the proof of Lemma 3.2, $\langle x, x\rangle + \langle v, v\rangle \equiv 0 \pmod 4$, and $\langle x, x\rangle + \langle v, v\rangle \leq 16$. The only pairs of even integers ≥ 6 with these properties are $(6, 6)$ with sum 12 and $(6, 10), (8, 8), (10, 6)$ with sum 16. □

In particular, the type $(6, 8)$, which would be the most frequent one by a naive count, is excluded by parity. Since $\tau^2 = -5 \equiv 1 \pmod 2$, the coset map $c \mapsto c\tau$ is an involution on $L_{56}/2L_{56}$, so a collision (x, v) of type $(10, 6)$ is the same as a collision (v, x) of type $(6, 10)$ (Appendix C.4).

Corollary 4.3. L_τ has no inner-shell vectors if and only if
(i) $\min(r\tau + 2L_{56}) \geq 14$ for every $r \in L_{56}(6)$ (the coset is odd, so its norms are $6, 10, 14, \ldots$; this says 6 and 10 do not occur), and
(ii) $\min(x\tau + 2L_{56}) \geq 12$ for every $x \in L_{56}(8)$ (the coset is even, so its norms are $8, 12, \ldots$; this says 8 does not occur).

(That the cosets in (ii) contain neither 0 nor vectors of norm 4 follows from the case $v = 0$ in the proof of Lemma 4.2 and from $\min L_{56} = 6$.)

4.2 Symmetry

Let $S = \mathrm{SL}_2(29) \subset \mathrm{Aut}(L_{56})$. Every $g \in S$ commutes with τ (exactly, not only modulo 2) [C2]. Hence the coset of $(xg)\tau$ is the image under g of the coset of $x\tau$, and g preserves norms. So condition (i) needs to be tested only for one representative of each S-orbit on $L_{56}(6)$, and (ii) for one representative of each S-orbit on $L_{56}(8)$.

To collect representatives for $L_{56}(8)$ we use the larger group $S^* = S \cdot C$. If $x_1, \ldots, x_K$ represent the S^*-orbits on $L_{56}(8)$, then the vectors $x_i u$ (u in a set of 6 representatives of $C/\{\pm 1\}$) represent all S-orbits (with repetitions). The elements of C do *not* commute with τ, which is why all pairs (x_i, u) must be tested.

Table 3. The cases to be tested.

shell	vectors	S-orbits	structure and test units
$L_{56}(6)$ (minimal vectors)	15 590 400	656	632 orbits of size 24 360 and 24 of size 8 120; 656 representatives
$L_{56}(8)$	36 957 286 800	1 517 130	all of size 24 360; 254 664 S^*-orbits $\times$ 6 elements of $C/\{\pm 1\} = 1\,527\,984$ pairs

4.3 Finite verification and the proof of Theorem A

Fix the data files of Appendix D: the Gram matrix G of L_{56}, the two generators of the symmetry group S, and the endomorphism matrix τ. The labels F0–F5 below identify the statements used by the verification programs; the headings describe their mathematical content.

Defining matrices (F0). The matrix G in Appendix D is positive definite, even integral and of determinant 1; (3.1) is also even and of determinant 1. [C2]

Minimum norm of the input lattice (F1). $\min L_{56} = 6$. [C5]

Complete norm-6 orbit list (F2). $L_{56}(6)$ is the disjoint union of the S-orbits of 656 explicitly listed vectors. [C8]

Complete norm-8 orbit list (F3). $L_{56}(8)$ is the union of the S-orbits of the 1 527 984 vectors $x_i u$; it consists of 1 517 130 regular S-orbits, that is, orbits with trivial stabilizer, hence of size $|S| = 24\,360$. [C9]

Symmetry and endomorphism identities (F4). The two generators of S preserve the Gram matrix, generate a group of order 24 360, and commute with τ; $\tau^2 = -5$ and $\tau G \tau^T = 5G$. [C2]

Lower bounds for the prescribed cosets (F5). (a) $\min(r\tau + 2L_{56}) \geq 14$ for each of the 656 representatives r. (b) $\min(x_i u\tau + 2L_{56}) \geq 12$ for each of the 1 527 984 pairs. Part (b) is established by two different methods (§§6.5 and 6.7); either one suffices. [C10, C11, C13]

Proof of Theorem A. The defining matrices give the even unimodular lattice of dimension 112 described in Lemma 3.2; the required endomorphism identities are part of F4. By F1 and Lemma 4.2, any nonzero vector of norm below 10 would give one of the four types listed in §1.3.

The isometries in S commute with τ by F4. Hence the minimum norm of the coset $(xg)\tau + 2L_{56}$ equals that of $x\tau + 2L_{56}$ for $g \in S$. By F2 and F3, the two lower bounds of Corollary 4.3 therefore need only be checked on the supplied norm-6 and norm-8 collections. F5 gives the bounds 14 and 12 on these collections. The type $(10, 6)$ is covered by the interchange argument of Appendix C.4. Thus no type can occur, and $\min L_{112} \geq 10$.

Since L_{112} is even unimodular of dimension 112, (1.1) gives $\min L_{112} \leq 10$. This proves the theorem. □

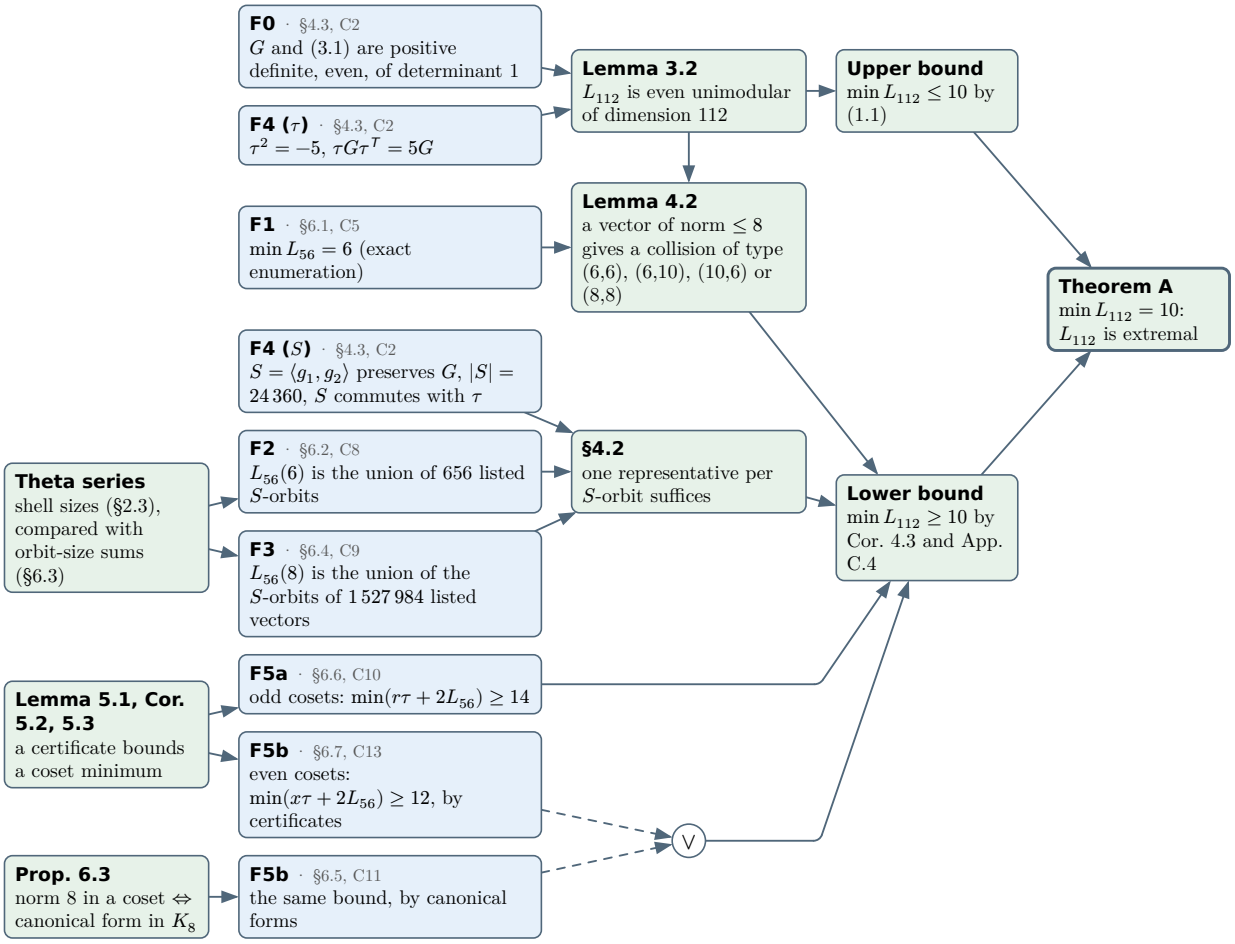


Figure 1. Dependencies in the proof of Theorem A. Blue boxes are the finite computations F0–F5 of §4.3, each labelled with the section that describes its algorithm and its row in Table D2; F4 appears twice, once for its identities about τ and once for those about S. Green boxes are statements proved in the text. An arrow from one box to another means that the second uses the first. F5b is established by two methods (§§6.5 and 6.7), either of which suffices ($\vee$). Consistency tests and supplementary arguments are not shown.

5. Certificates for coset minima

We derive lower bounds for a coset minimum from the norm of one representative and its inner products with all minimal vectors. The midpoint lemma proves that the stated conditions on a representative imply the required bound; the finite computations later check these conditions. A representative w_0 satisfying these conditions (Check 5.4) is called a **certificate** for its coset: checking it is a finite computation, and the corollaries below turn it into a proof of the bound.

Lemma 5.1 (midpoint lemma). Let L be even unimodular with $\min L = 6$. If $w, w' \in L$ lie in the same coset modulo $2L$ and $w \neq \pm w'$, then

$$\langle w, w\rangle + \langle w', w'\rangle \pm 2\langle w, w'\rangle \geq 24 \qquad \text{for both signs.}$$

Proof. $w \mp w' \in 2L \setminus \{0\}$, so the midpoint $\frac{1}{2}(w \mp w')$ is a nonzero vector of L and has norm ≥ 6; multiply by 4. □

The two corollaries below retain the assumptions on L in Lemma 5.1. Two immediate consequences follow. Two vectors of norm 6 that are distinct up to sign never lie in the same coset (with $t = \langle w, w'\rangle$,

the inequalities $12 + 2t \geq 24$ and $12 - 2t \geq 24$ cannot both hold), and the same holds for norm 8. Hence each coset contains at most one pair $\pm w$ of norm 6, and at most one pair of norm 8 (Appendix C.3).

Corollary 5.2 (lower bound for odd cosets). If an odd coset c contains a vector w_0 of norm $N \in \{14, 18\}$ such that $|\langle w_0, u\rangle| \leq 6$ for every $u \in L(6)$, then $\min(c) \geq 14$.

Proof. Suppose $w \in c$ has norm $m \leq 10$. Norms in c are $\equiv N \pmod 4$, so $m \in \{6, 10\}$; in particular $m < N$, $N + m \leq 28 < 32$ and $w \neq \pm w_0$. The midpoints $a = \frac{1}{2}(w_0 + w)$ and $b = \frac{1}{2}(w_0 - w)$ are nonzero vectors of L (Lemma 5.1) with $\langle a, a\rangle + \langle b, b\rangle = \frac{1}{2}(N + m) < 16$. Since L is even with minimum 6, they cannot both have norm ≥ 8; so one of them is a minimal vector u, and $w_0 - 2u = \pm w$. Taking norms, $N - 4\langle w_0, u\rangle + 24 = m$, i.e. $\langle w_0, u\rangle = \frac{1}{4}(N + 24 - m) > 6$, a contradiction. □

Corollary 5.3 (lower bound for even cosets). If an even coset c contains a vector w_0 of norm $N \in \{12, 16, 20\}$ such that $|\langle w_0, u\rangle| \leq 6$ for every $u \in L(6)$, then $\min(c) \geq 12$.

Proof. Norms in the even coset c are $\equiv 0 \pmod 4$. The coset does not contain 0 (otherwise $w_0 \in 2L$ and $N \geq 4 \cdot 6$), nor a vector of norm 4 (since $\min L = 6$). If $w \in c$ had norm $m = 8$, then $m < N$ and $N + m \leq 28$, and the argument of Corollary 5.2 gives a minimal vector u with $\langle w_0, u\rangle = \frac{1}{4}(N + 16) > 6$, a contradiction. □

Check 5.4 (coset-bound data). To verify a lower bound for a coset c, supply a representative $w_0 \in c$ satisfying Corollary 5.2 or 5.3. Check:
(a) $w_0 \in c$; when $c = x\tau + 2L$, every coordinate of $w_0 - x\tau$ is even.
(b) $\langle w_0, w_0\rangle$ belongs to $\{14, 18\}$ for an odd coset, or to $\{12, 16, 20\}$ for an even coset.
(c) $|\langle w_0, u\rangle| \leq 6$ for every $u \in L(6)$, using the complete minimal-vector list.

These conditions are sufficient but not necessary: a coset without short vectors need not have a representative satisfying them. For every coset required in §§6.6–6.7, such a representative is supplied and checked.

6. Algorithms and the verification

Two implementations are referred to below. Implementation 1 constructed L_{112} and performed the main computations; its checks, producers and library are `code/verify/`, `code/produce/` and `code/lib/` of [Repo]. Implementation 2 re-implemented the checks without reading the code of implementation 1, sharing with it the data files and the verification data; it is `code/crosscheck/`. Both were written by sessions of the same AI system (see "Use of AI tools"); what they share is listed in Supplement §S3.3. For each computational fact we describe the algorithm and the checks on its output. Completeness requires both disjointness and agreement of the orbit-size sum with the theta-series coefficient. The two rigorous even-coset methods share these orbit lists and the odd-coset bounds (§1.3); separate implementations and tests on gluings A and B check their results. Floating-point searches and samples have the supplementary scope stated below. Running times are elapsed times of single-threaded runs on the machine used (Intel Core Ultra 9 288V, 8 cores, 15 GB of memory, Linux under WSL2, shared with other jobs) unless stated otherwise; they vary with the load (Supplement §S3.1). Code references are given in Appendix D and pseudocode in Appendix E.

Table 4. The computational facts and how they were checked. The main methods, and the other methods marked "exact", are proofs; the remaining entries are supporting evidence. The design identities are sampled consistency checks in exact arithmetic, not proofs of completeness.

fact	main method	other methods and separately written reproductions	nature
F1	interval-arithmetic enumeration (BKZ-40 basis)	exact: integer enumeration (a′). Reduction to $L_{56,2}(M)$ plus $\mathbb{F}_2$ linear algebra (assumes the literature value of $\min L_{56,2}(M)$); fplll (two bases) and PARI `qfminim` (floating point)	interval arithmetic; (a′) integer
F2	closure + orbits, size sum	exact: expansion and disjointness (implementation 2). Sampled consistency check in exact arithmetic: design identities (implementation 2)	integer
F3	marked sums of two minimal vectors, size sum	exact: recount from K_8. Sampled consistency check in exact arithmetic: design identities (implementations 1 and 2)	integer
F4	matrix identities	exact: implementation 2	integer
F5a	656 odd-coset certificates	exact: the checker of implementation 2; canonical forms. CVP on two bases (floating point)	integer
F5b	canonical forms (two implementations)	exact: even-coset certificates (all pairs). Exhaustive CVP (floating point); a sample of 180 pairs (implementation 2)	integer (CVP: floating point)

6.1 F1: the minimum of the base lattice

The enumeration (a), in interval arithmetic, and the enumeration (a′), in integer arithmetic, cannot miss vectors; either one proves F1 directly from the data file without depending on the minimum reported in the literature (§2.4). Route (b), by contrast, assumes the literature value and is used to check consistency with it; (c) is a supplementary check in floating-point arithmetic.

(a) Interval enumeration. [C5]

1. Reduce the Gram matrix with BKZ-40. Check exactly that the transformation U is an integer matrix with $\det U = \pm 1$ and compute $G' = UGU^T$ in integers (floating point used inside the reduction does not affect correctness; U is distributed as `certificates/reduction_L.txt`).
2. Compute $G' = V^T \operatorname{diag}(D) V$ (V unit upper triangular) **exactly in rational arithmetic**.
3. Replace every entry q of D and V by the interval whose ends are the two doubles adjacent to the double nearest to q; that q lies in it is checked in rational arithmetic.
4. Run Fincke–Pohst enumeration [FP85, SE94] in interval arithmetic, and prune only when the **lower end** of the interval of the partial norm exceeds the radius 4.

Rounding model. The program uses IEEE binary64 arithmetic with round-to-nearest and never changes the rounding mode. After every operation the computed result a is replaced by the interval $[a -$

$|a|\ 2^{-51} - 2^{-1060}, a + |a|\ 2^{-51} + 2^{-1060}]$. Round-to-nearest has relative error at most 2^{-53} for normal results and absolute error at most 2^{-1075} for subnormal ones, so this interval contains the exact result of the operation on the computed operands; the absolute term also covers results that are 0 or close to 0. The ends of a product of intervals are the minimum and the maximum of the products of the ends, and the lower bound of a square is 0 when the interval contains 0. The program is compiled with `-O2 -fno-fast-math -ffp-contract=off`, so that every operation is rounded separately (no fused multiply-add); the analysis assumes evaluation in binary64 without extended precision, as on x86-64 with SSE2 and on arm64 (we used gcc 15.3 on x86-64).

Pruning. At each level the admissible values of the coordinate lie in an interval computed from an upper bound of the remaining budget and from the interval of the centre; this interval is enlarged by one integer on each side, and every integer in it is visited (there is no zig-zag stopping rule, so no monotonicity argument is needed). A branch is cut only when the lower end of the new partial norm exceeds 4. The true partial norm always lies in its interval, so no vector of true norm ≤ 4 can be pruned. Only one of $\pm x$ is enumerated, and the norms of the vectors reaching the leaves are computed exactly with 128-bit integers. The results are in Table 5: for L_{56}, 8 454 837 756 nodes were visited and the only leaf reached was the zero vector, which is reached once and not counted (34 minutes with the published code).

Table 5. Results of the interval enumeration (only one of $\pm x$ is enumerated; counts include both signs). The last two controls have different roles. In skewed bases the entries are large, and an enumerator without the outward rounding loses vectors of E_8 there, while it passes the controls in reduced bases. The random positive definite forms ($G = BB^T + I$, entries of B in $\{-3, ..., 3\}$) are compared with exhaustive search; they test vectors of norm equal to the radius and the sign rule, and did not detect the removal of the outward rounding.

lattice	dim	radius	nodes	nonzero vectors found	expected
E_8 (control)	8	2	248	240 of norm 2	240
Leech lattice (control)	24	4	2 019 561	196 560 of norm 4, none of norm 2	same
$\Lambda_{24} \perp \Lambda_{24} \perp E_8$ (control)	56	4	14 101 914	240 of norm 2, 395 280 of norm 4	same
$\boldsymbol{L_{56}}$	56	4	8 454 837 756	**none**	—
E_8 in 5 skewed bases (control)	8	4	—	240 of norm 2, 2 160 of norm 4 in each basis	same
60 random positive definite forms (control)	2–5	1–14	—	230 vectors (115 $\pm$-pairs), 44 of them of norm equal to the radius	exhaustive search

(a′) A second exact enumeration, by a different method. A separately written program (developed without reference to the code of (a)) makes every decision in integer arithmetic instead of interval arithmetic. The approximate Cholesky factor of G' (BKZ-40 basis, U checked exactly as in (a)) is scaled by 2^{40} and rounded to an integer upper triangular matrix R; then $Q(x) = |Rx^T|^2$ is an integer quadratic form, the error matrix $E = 2^{80}G' - R^TR$ is computed exactly, and $|xEx^T| \leq \|E\|_F \left\|G'^{-1}\right\|_\infty \langle x, x\rangle$ with G'^{-1} an exact integer matrix (since $|xEx^T| \leq \|E\|_2\ |x|^2$ and $|x|^2 \leq \left\|G'^{-1}\right\|_2 \langle x, x\rangle$, where $\|E\|_2 \leq \|E\|_F$, and $\left\|G'^{-1}\right\|_2 = \rho(G'^{-1}) \leq \left\|G'^{-1}\right\|_\infty$ for the symmetric matrix G'^{-1}). Hence $\langle x, x\rangle \leq 4$ implies $Q(x) \leq T := 4 \cdot 2^{80} + 4\|E\|_F \left\|G'^{-1}\right\|_\infty$ (a relative enlargement of 3.8×10^{-7}). All x with $Q(x) \leq T$ are enumerated in C with 128-bit integers, the range at each level being given by exact integer square roots, so no vector is pruned because of rounding. No overflow can occur: the program checks, and stops

otherwise, that $T \le 2^{120}$, that every entry of R is at most 2^{80} and every coordinate at most 2^{40} in absolute value, and that $|y| \le 2^{61}$ before a value y is squared; then every intermediate value is below $2^{122} < 2^{127}$ in absolute value. These are explicit tests, not assertions, so they do not depend on compiler flags. For L_{56}: 14 773 398 896 nodes (both signs), no nonzero vector found (96 minutes) [C27]. Positive control: E_8^7 in a random basis gives 1 680 vectors of norm 2 and 1 224 720 of norm 4, as expected.

(b) Reduction to Nebe's lattice. L_{56} is the sublattice $L_{56,2}(M)X + 2L_{56,2}(M)$ of $L_{56,2}(M)$ (where X is an element of $M/2M$ whose action modulo 2 has an image of dimension 28), rescaled by $1/\sqrt{2}$; the Gram matrix of this sublattice agrees entry by entry with that of L_{56}. Hence norm-4 vectors of L_{56} correspond bijectively to norm-8 vectors of $L_{56,2}(M)$ whose class lies in an $\mathrm{SL}_2(29)$-invariant 28-dimensional subspace W of $L_{56,2}(M)/2L_{56,2}(M)$, and norm-2 vectors of L_{56} to norm-4 vectors of $L_{56,2}(M)$. Consider the 1 528 956 pairs (norm-8 orbit representative of $L_{56,2}(M)$, centralizer element); the 127 413 representatives have orbit sizes summing to the theta coefficient. None of these pairs has its coset in W [C6]. Positive control: the 656 orbit representatives of minimal vectors of L_{56} all appear in $L_{56,2}(M)$ with norm 12 and class in W. This proves "$\min L_{56,2}(M) = 6 \Rightarrow \min L_{56} = 6$", reducing F1 to the minimum of $L_{56,2}(M)$, the entry of [Neb98, Table II] whose minimal vectors were enumerated (and which is checked again in [C6]).

(c) Floating-point enumerations. fplll enumeration of all vectors of norm ≤ 4.5 (two bases) and PARI `qfminim` for norm ≤ 4 (BKZ-50 basis) also found none [C7].

6.2 F2: orbits of minimal vectors

We grow the set of minimal vectors by a closure operation, split it into orbits, and stop when the orbit sizes add up to the kissing number 15 590 400. Disjointness and agreement of the total sizes prove completeness.

Starting from a few minimal vectors found by BKZ, we generated minimal vectors by closure: if $r, s \in L_{56}(6)$ and $\langle r, s\rangle = 3$ then $r - s \in L_{56}(6)$. For each new vector we computed its S-orbit (by applying all 12 180 elements of $S/\{\pm 1\}$) and recorded it as one orbit; the closure step produces vectors outside the orbits already found. We obtained 656 distinct orbits, with sizes adding up to $632 \times 24\,360 + 24 \times 8\,120 = 15\,590\,400 = |L_{56}(6)|$ [C8]. Since distinct orbits are disjoint, **completeness follows from the size sum and F1**. Implementation 2 expanded all orbits, checked pairwise disjointness, and checked the identities of §6.3 for $m = 6$ [C8].

6.3 Completeness tests: orbit-size sums and design identities

In addition to the size sum, we verified exact identities (equalities of integers of size about 10^{18}) that follow from the shells being spherical 7-designs, computed from the lists of representatives. Barring coincidence, duplicates or omissions would violate them.

For every $a \in \mathbb{R}^{56}$

$$\sum_{x \in L_{56}(m)} \langle x, a\rangle^2 = |L_{56}(m)| \frac{m\langle a, a\rangle}{56},$$

$$\sum_{x \in L_{56}(m)} \langle x, a\rangle^4 = |L_{56}(m)| \frac{3m^2\langle a, a\rangle^2}{56 \cdot 58},$$

$$\sum_{x \in L_{56}(m)} \langle x, a\rangle^6 = |L_{56}(m)| \frac{15m^3\langle a, a\rangle^3}{56 \cdot 58 \cdot 60}.$$

The left sides were computed from the representative lists (representative × group element × element of C, weighted by orbit sizes) for three integer vectors a and for $m = 6$ and $m = 8$; all 18 identities hold exactly [C8]. Implementation 2 computed them separately, for $m = 6$ from the expanded orbits and for $m = 8$ from the S^*-orbits of the representatives [C8, C9].

6.4 F3: orbits of norm-8 vectors via marked sums of two minimal vectors

The approximately 37 billion vectors are too many to store, so we distinguish orbits by marking only those of their elements that are sums of two minimal vectors. Completeness is again proved by the size sum.

Every norm-8 vector we need can be written as $x = r + s$ with $r, s \in L_{56}(6)$ and $\langle r, s\rangle = -2$, and by symmetry r may be taken among the (at most 150) S^*-orbit representatives r_i of minimal vectors. Let $X_i = \{r_i + s : s \in L_{56}(6), \langle r_i, s\rangle = -2\}$.

> **Definition 6.1.** For an S^*-orbit $O \subset L_{56}(8)$, the **marked elements** of O are the elements of $O \cap \bigcup_i X_i$, that is, the elements of O that are sums $r_i + s$ of the above shape (possibly in several ways). Completeness of the orbit list does not rely on every norm-8 vector being such a sum: it follows from the orbit-size sum (Table 3).

The algorithm runs through candidates $x \in X_i$; whenever x is not yet marked, it starts a new orbit, computes all elements of that orbit in $\bigcup_i X_i$ and marks them. This needs only one matrix product per orbit: $y = \pm xug^{-1}$ lies in X_i iff $\langle y, r_i\rangle = 4$, i.e. iff $\langle xu, r_i g\rangle = \pm 4$, so precomputing $W = \{r_i g\}$ finds them all at once. The decompositions of $x = r_i + s_0$ can be read off from the minimal vectors with inner product 2 or 3 with r_i, which reduces the work by a factor of about 30. The stabilizer order of x is the number of elements of S^* that map x to itself, read off from the same matrix product (they are group elements, not decompositions); a wrong count would make the orbit sizes, and hence their sum, wrong. Only the marked elements (tens of millions) are stored.

The result is Table 3: $1\,517\,130 \times 24\,360 = 36\,957\,286\,800 = |L_{56}(8)|$ [C9]. This was checked in three ways: (1) the S^*-orbit sizes add up to $|L_{56}(8)|$; (2) the set K_8 of canonical forms (§6.5) of the cosets of all $x_i u$ has exactly $1\,517\,130$ elements, and recomputing each orbit size from K_8 gives the same total (which would fail if an orbit were counted twice); (3) the identities of §6.3. The first 25 orbits were cross-checked against a naive method, and the canonical forms of the cosets of $2\,000$ norm-8 vectors $a \mp b$, formed without the orbit list from random minimal vectors a, b with $\langle a, b\rangle = \pm 2$, were all found in K_8 [C34].

Which computed facts prove F2 and F3. By Appendix C.3, $x \mapsto x + 2L_{56}$ is injective up to sign on $L_{56}(6)$ and on $L_{56}(8)$ (this uses F1). Hence two vectors of norm 8 lie in the same S-orbit if and only if their cosets do, and the S-orbit of x has twice as many elements as the S-orbit of $x + 2L_{56}$ in $L_{56}/2L_{56}$, on which $S/\{\pm 1\}$ acts. The proof of F3 (step v4) uses exactly this: the $1\,517\,130$ distinct canonical forms (§6.5) of the cosets of the $x_i u$ give pairwise distinct S-orbits, the size of each orbit is computed from its coset as $2 \cdot 12\,180 = 24\,360$ (so every orbit is regular; the stabilizers in S^* computed by the producer are not used), and the sizes add up to $|L_{56}(8)|$. Distinctness and sizes therefore rest on the canonical-form code for every route of Supplement Table S7, including route B, which avoids canonical forms only in the test for F5b. They are re-checked without canonical forms modulo 2 by implementation 2, which names an orbit by its lexicographically smallest vector [C27]. For F2, step v3 checks distinctness and

sizes both by expanding the 656 orbits and by canonical forms, and implementation 2 expands them again [C8].

6.5 F5b: canonical forms on $L_{56}/2L_{56}$

A coset is stored as a 56-bit integer and its $\mathrm{SL}_2(29)$-orbit is named by the smallest element (canonical form). Since a class contains a norm-8 vector if and only if its canonical form lies in the table K_8, each test is a single table look-up. Only integer arithmetic is used, so no rounding occurs.

Identify $L_{56}/2L_{56}$ with $\mathbb{F}_2^{56}$ via coordinates modulo 2. Each $g \in S/\{\pm 1\}$ acts by an $\mathbb{F}_2$-linear map, implemented with byte-wise lookup tables and XOR.

Definition 6.2 (canonical form). The **canonical form** $\mathrm{can}(c)$ of a coset c is the smallest 56-bit integer in its orbit $\{cg : g \in S\}$.

Proposition 6.3. Let $K_8 = \{\mathrm{can}(x_i u + 2L_{56})\}$ (all pairs). Assuming F3, a coset c contains a vector of norm 8 if and only if $\mathrm{can}(c) \in K_8$.

Proof. If $w \in c$ has norm 8, by F3 $w = x_i u g$ for some i, u and $g \in S$, so $c = (x_i u + 2L_{56})g$ and $\mathrm{can}(c) = \mathrm{can}(x_i u + 2L_{56}) \in K_8$. Conversely, $\mathrm{can}(c) = \mathrm{can}(x_i u + 2L_{56})$ means $c = (x_i u + 2L_{56})g$ for some g, and $x_i u g \in c$ has norm 8. □

Test for type $(8,8)$: for each pair, compute the coset of $x_i u \tau$, its canonical form (12 180 table look-ups, about 0.3 ms), and look it up in K_8. The results are in Table 6. The test was implemented twice with separately written code (group enumeration, tables, canonical forms, look-up), with self-tests (table look-up equals matrix product; canonical form is S-invariant; the 12 180 group elements are distinct modulo 2). The type $(6,6)$ was tested on the 656 minimal-vector cosets using $K_6 = \{\mathrm{can}(r_i + 2L_{56})\}$. The first row of Table 6 is a positive control on a sample [C34], and implementation 2 also tested a sample of 180 pairs by enumerating the lattice $\langle 2L_{56}, t\rangle$ instead of using canonical forms (Supplement §S3.3).

Table 6. Results of the canonical-form test for type $(8,8)$ (identical for τ and τ^{-1} mod 2).

gluing	pairs tested	colliding pairs	colliding S-orbits	norm-8 vectors of L_f
class containing an element of C	3 000 (sample)	3 000	—	— (minimum 6)
gluing A	1 528 956	36	6	292 320
gluing B	1 527 984	10	6	292 320
gluing C	1 527 984	**0**	**0**	**0**

6.6 F5a: certificates for the types $(6,6)$, $(6,10)$, $(10,6)$

All 656 odd cosets have a certificate of norm 14 or 18. The search is heuristic; its output is checked by Check 5.4.

For each representative r we start from $w = r\tau$ (norm 30) and, while some minimal vector u has $\langle w, u\rangle \geq 7$, replace w by $w - 2u$ (the **descent**). This lowers the norm by $4(\langle w, u\rangle - 6) > 0$ and stays in the class. Positive control: starting from the sum of a norm-10 vector and an element of $2L_{56}$, the descent ended at norm 10 in all 60 trials and never produced a false certificate.

Table 7. The 656 odd-coset certificates.

$\langle w_0, w_0\rangle$	number	maximum of $\lvert\langle w_0, u\rangle\rvert$ over all minimal vectors
14	243	5 (43 vectors), 6 (200)
18	413	6 (all 413)

The certificates were produced by one implementation and re-checked by a separate implementation with separately written code, which also re-derived the lemma [C10]. Two closest-vector enumerations on different bases independently found no vector of norm ≤ 10 in these classes [C12].

6.7 F5b: bounds from even-coset representatives

Even-coset certificates. For every pair we computed an even-coset certificate (Corollary 5.3) by the descent of §6.6, starting from $w = x_i u\tau$ (norm 40). Inner products are computed in float64; they are exact because every partial sum of a product AB of integer matrices is bounded by $\sum_k |a_k|\,|b_k| \leq \left(\max_{\text{row}} \sum_k |a_k|\right) \cdot \max|B|$, which is checked to be below 2^{52} at run time for both products used. The bound holds for every order of summation and with or without fused multiply-add, so the result does not depend on how the BLAS library (OpenBLAS 0.3.34 in our runs) arranges the sums; it assumes only that the library forms sums of the individual products in IEEE arithmetic, not Strassen-type or reduced-precision products. The result: **every one of the 1 527 984 pairs has a certificate** (Table 8) [C13]. Thus F5b also holds by an argument that uses only the midpoint lemma, integer inner products and the list F2, independently of canonical forms. All certificates were re-checked by a separately written program, using integer arithmetic and float32 products only where all partial sums are proved to be below 2^{24} [C27].

Table 8. Even-coset certificates for all 1 527 984 pairs.

$\langle w_0, w_0\rangle$	max $\lvert\langle w_0, u\rangle\rvert$	number	share
12	5	41 437	2.71 %
12	6	692	0.05 %
16	6	1 465 524	95.91 %
20	6	20 331	1.33 %

7. Identification and formalization

7.1 Formalization of the deduction (Lean)

The deduction from the computational facts to Theorem A has been proved in the proof assistant Lean 4 (with mathlib) and machine-checked. The Lean statement (`theoremA` and `theoremA_canonical` in `lean/EEUL/L112/Chain.lean`) is: **assuming the evenness part of F0 and F1–F5, if $(x, x\tau + 2y) \neq 0$ then $\langle x, x\rangle + \langle x\tau + 2y, x\tau + 2y\rangle \geq 20$.** By (1.2) this says that L_{112} has no inner-shell vector. The evenness part of F0 and the facts F1–F5 are hypotheses of the Lean theorems, and the translation through (1.2) is outside Lean. The formalization excludes errors that consist of a gap in this deduction relative to the stated hypotheses (a missing type, a missing case, or a wrong direction).

The setting is abstract: L_{56} is an additive group with an integer-valued symmetric bilinear form B, even norms, nonzero vectors of norm ≥ 6 (F1), and τ with $B(x\tau, y\tau) = 5B(x, y)$ and $\tau^2 = -5$ (F4). The symmetry group, the representatives and the certificates are the fields of a structure whose axioms are F2–F5. Lean writes maps on the left (τx and gx for our $x\tau$ and xg). The two ways of checking F5b enter as two forms of the hypothesis: "the coset of each norm-8 representative contains no vector of norm 8" (canonical forms, §6.5; Proposition 6.3, which turns the computed $\mathrm{can}(c) \notin K_8$ into this statement, is applied outside Lean) and "the coset of each norm-8 representative has an even certificate" (§6.7, Corollary 5.3). The formalized statements are listed in Table 9. The hypotheses and the two main

statements read as follows in Lean (proofs and comments omitted); Table 9 (continued) gives for each hypothesis the check that establishes it.

Table 9. Statements proved in Lean (`lean/EEUL/L112/Chain.lean`).

Lean theorem	in this paper	content
`no_inner_shell_ne8`, `no_inner_shell`	Lemma 4.2, Corollary 4.3, Appendix C.4	if (i) $\min(r\tau + 2L_{56}) \geq 14$ for every $r \in L_{56}(6)$ and (ii) for every $x \in L_{56}(8)$ the coset $x\tau + 2L_{56}$ contains no vector of norm 8 (or $\min(x\tau + 2L_{56}) \geq 12$), then $\langle x, x\rangle + \langle x\tau + 2y, x\tau + 2y\rangle \geq 20$ whenever $(x, x\tau + 2y) \neq 0$
`midpoint_lemma`	Lemma 5.1	if $w \neq \pm w'$ lie in the same coset, then $\langle w, w\rangle + \langle w', w'\rangle \pm 2\langle w, w'\rangle \geq 24$
`cert_bound`, `odd_bound`, `even_bound`	Corollaries 5.2, 5.3	a certificate (Check 5.4) of norm N with $12 \leq N \leq 20$ forces all vectors of its coset to have norm ≥ 12; hence ≥ 14 for odd cosets ($N \in \{14, 18\}$) and ≥ 12 for even cosets ($N \in \{12, 16, 20\}$)
`class_pred_transport`, `class_bound_transport`	§4.2	a property of the norms in the coset $r\tau + 2L_{56}$ of a representative holds in the coset $(rg)\tau + 2L_{56}$ for every $g \in S$
`theoremA`	§4.3, derivation of Theorem A with F5b checked by certificates (§6.7)	assuming F1 and F2–F5 (the elements of S are isometries commuting with τ; the norm-6 and norm-8 vectors are images of the representatives; the coset of each norm-6 representative has an odd certificate, that of each norm-8 representative an even one), $\langle x, x\rangle + \langle x\tau + 2y, x\tau + 2y\rangle \geq 20$ whenever $(x, x\tau + 2y) \neq 0$
`theoremA_canonical`	§4.3, derivation of Theorem A with F5b checked by canonical forms (§6.5)	the same conclusion, with F5b in the form "the coset of each norm-8 representative contains no vector of norm 8"

```
structure Setting (L : Type*) [AddCommGroup L] where
  B : L →+ L →+ ℤ
  symm : ∀ x y, B x y = B y x
  even : ∀ x, B x x % 2 = 0
  min6 : ∀ x, x ≠ 0 → 6 ≤ B x x
  τ : L →+ L
  τ_sim : ∀ x y, B (τ x) (τ y) = 5 * B x y
  τ_sq : ∀ x, τ (τ x) = -(5 • x)

def N (x : L) : ℤ := S.B x x
def SmallIP (w0 : L) : Prop := ∀ u, S.N u = 6 → -6 ≤ S.B w0 u ∧ S.B w0 u ≤ 6
```

```
structure Data (S : Setting L) where
  Sym : Set (L ≃+ L)
  reps6 : Set L
  reps8 : Set L
  sym_iso : ∀ g ∈ Sym, ∀ a b, S.B (g a) (g b) = S.B a b
  sym_tau : ∀ g ∈ Sym, ∀ a, S.τ (g a) = g (S.τ a)
  cover6 : ∀ x, S.N x = 6 → ∃ g ∈ Sym, ∃ r ∈ reps6, x = g r
  cover8 : ∀ x, S.N x = 8 → ∃ g ∈ Sym, ∃ r ∈ reps8, x = g r
  bound6 : ∀ r ∈ reps6, ∃ w0 y0, w0 = S.τ r + 2 • y0 ∧ (S.N w0 = 14 ∨ S.N w0 = 18) ∧ SmallIP S w0

structure DataCert (S : Setting L) extends Data S where
  bound8 : ∀ r ∈ reps8, ∃ w0 y0, w0 = S.τ r + 2 • y0 ∧ (S.N w0 = 12 ∨ S.N w0 = 16 ∨ S.N w0 = 20) ∧
    SmallIP S w0

structure DataCanonical (S : Setting L) extends Data S where
  nocoll8 : ∀ r ∈ reps8, ∀ y, S.N (S.τ r + 2 • y) ≠ 8

theorem theoremA (D : DataCert S) (x y : L) (hz : x ≠ 0 ∨ S.τ x + 2 • y ≠ 0) :
    20 ≤ S.N x + S.N (S.τ x + 2 • y)
theorem theoremA_canonical (D : DataCanonical S) (x y : L) (hz : x ≠ 0 ∨ S.τ x + 2 • y ≠ 0) :
    20 ≤ S.N x + S.N (S.τ x + 2 • y)
```

Table 9 (continued). The hypotheses of `theoremA` and `theoremA_canonical` and the checks that establish them. Steps v0–v6 are run by `code/verify/proof.sh`, v7 by `full.sh`. Checks in monospace are printed by the programs and recorded in `out/<step>.json`.

Lean hypothesis	fact	checked by
`Setting.symm`, `Setting.even`	F0 (evenness)	v0: `G is symmetric with even diagonal`
`Setting.min6`	F1	v1: no nonzero vector of norm ≤ 4, after the exact check of the basis change; with v0 (`G is positive definite`) and evenness, every nonzero norm is ≥ 6
`Setting.τ_sim`, `Setting.τ_sq`	F4	v0: `tau G tau^T = 5 G, tau^2 = -5 I`
`Data.sym_iso`, `Data.sym_tau`	F4	v0: `g1 G g1^T = G, g1 tau = tau g1` and the same for g_2; the extension to all of S is outside Lean (see below)
`Data.reps6`, `Data.cover6`	F2	v3: the 656 orbits are distinct and their sizes add up to $\lvert L_{56}(6)\rvert$ (from v2); the conclusion uses F1 and the theta series (outside Lean)
`Data.reps8`, `Data.cover8`	F3	v4: `exactly 1517130 distinct S-orbits, all orbits are regular (size 24360), sizes add up to |L_8| = 36957286800`, with v2
`Data.bound6`	F5a	v5: Check 5.4 (a)–(c) for the 656 representatives
`DataCert.bound8`	F5b	v7: Check 5.4 (a)–(c) for the 1 527 984 pairs
`DataCanonical.nocoll8`	F5b	v6: no class $x_i u_k \tau + 2L_{56}$ has its canonical form in K_8 (built by v4); Proposition 6.3 (outside Lean)
"every u with `S.N u = 6`" in `SmallIP`	Check 5.4 (c)	v5, v7: the inner products are taken with all minimal vectors, expanded from the list of F2 (`code/lib/shells.py`)

Check: the package builds without warnings (a `sorry` anywhere would give one), each theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, and every declaration passes the kernel again when replayed by `leanchecker`. The script `lean/check.sh` runs the three checks; the expected axioms are in `lean/expected-axioms.txt` [C22].

Not formalized (outside the Lean statements):

- the computational facts themselves: F0 (unimodularity, an integer determinant), the matrix identities of F4, and the individual numbers in F1–F5 (§6);

- the passage from the pair $(x, x\tau + 2y)$ to the vector $(x, x\tau + 2y)/\sqrt{2} \in L_{112}$ and the factor $1/2$ in its norm (1.2);
- the extension of F4 from the two generators checked by the programs to the whole group S (the Lean hypothesis is about every element of S);
- the completeness arguments behind F2 and F3 (disjoint orbits whose sizes add up to the theta-series coefficient, §6.2 and §6.4) and the theta series itself (§2.3, Appendix B); Lean assumes the conclusion "every vector of norm 6 (norm 8) is the image of a representative";
- Proposition 6.3, which turns the computed $\mathrm{can}(c) \notin K_8$ into "the coset c contains no vector of norm 8";
- the upper bound (1.1), $\min L_{112} \leq 10$.

7.2 Identification with Nebe's lattice

Our L_{56} is $\mathrm{SL}_2(29)$-equivariantly isometric to $L_{56,2}(\widetilde{M})$ built from Nebe's definition [Neb98] without external data [C23].

1. Build $L_{56'} = L_{56,2}(\widetilde{M})$ and generators z' (order 29), w' (order 4) of $\mathrm{SL}_2(29)$ from the model over $\mathbb{Q}(\zeta_{29})$ (code: `code/supplementary/nebe/cycloquat.py`; of the two classes of maximal orders, $\widetilde{M}$ is the one containing a maximal order of $\mathcal{Q}_{\infty,3}$ [Neb98, p. 483], equivalently the one whose units of reduced norm 1, and hence the centralizer, form a group of order 12; §2.4, [C33]).
2. Find a group isomorphism between the two copies of $\mathrm{SL}_2(29)$ with GAP (also composed with the outer automorphism), and compute the space of equivariant maps $L_{56'} \to L_{56}$; it is 8-dimensional.
3. On its integral part (rank 8) enumerate the elements with $Q(T) = \mathrm{tr}(G'^{-1}TGT^T) \leq 56$; isometries are exactly the elements with $Q = 56$.

We find 6 (up to sign) unimodular integer matrices T with $TGT^T = G'$ intertwining the two actions, as many as predicted by the centralizer of order 12. Moreover, in the coordinates of $L_{56'}$, τ becomes right multiplication by an element of $\widetilde{M}$ of reduced norm 5. Thus **gluing C is built from Nebe's lattice $L_{56,2}(\widetilde{M})$ and an element of reduced norm 5 and reduced trace 0 of its maximal order $\widetilde{M}$; such elements fall into two classes modulo 2 in the sense of §3.3, and τ belongs to the class giving minimum 10**.

The element explicitly. `cycloquat.py` works in the algebra $\mathbb{Q}(\zeta_{29}) \oplus \mathbb{Q}(\zeta_{29})b$ with $b^2 = -1$ and $bx = \bar{x}b$, where $\bar{x}$ is the complex conjugate; The Borel subgroup B of $\mathrm{SL}_2(29)$ (of order $29 \cdot 28$) acts on it from the left as in [Neb98, Lemma 3.1, Prop. 3.2]; the element w of order 4 that generates $\mathrm{SL}_2(29)$ together with B is taken from the Weil representation. This explicit model, a concrete form of [Neb98, Rem. 3.3], is ours. Let $K_4 \subset \mathbb{Q}(\zeta_{29})$ be the quartic subfield; its ring of integers has the $\mathbb{Z}$-basis of Gaussian periods $\eta_j = \sum_{k=0}^{6} \zeta_{29}^{2^{4k+j}}$ ($j = 0, ..., 3$; 2 is a primitive root modulo 29). Complex conjugation exchanges $\eta_0 \leftrightarrow \eta_2$ and $\eta_1 \leftrightarrow \eta_3$, and its fixed field in K_4 is $\mathbb{Q}\left(\sqrt{29}\right)$. The commutant of $\mathrm{SL}_2(29)$ is right multiplication by $\mathcal{Q} = K_4 \oplus K_4 b$, the quaternion algebra of §2.4. The maximal order $\widetilde{M}$ constructed by the program is the $\mathbb{Z}$-module with basis

$$m_0 = \frac{\eta_0 + 41\eta_1 + 144\eta_2 + 104\eta_3 + (133\eta_0 + 88\eta_1 + 12\eta_2 + 492\eta_3)b}{145},$$

$$m_1 = \eta_1 + 2\eta_3 b, \quad m_2 = \eta_2 + 3\eta_3 b, \quad m_3 = \eta_3 + 3\eta_3 b,$$

$$m_4 = (\eta_0 + \eta_3)b, \quad m_5 = (\eta_1 + \eta_3)b, \quad m_6 = (\eta_2 + 4\eta_3)b, \quad m_7 = 5\eta_3 b.$$

In the coordinates of $L_{56'}$, the endomorphism τ is right multiplication by

$$h = x + yb, \quad x = \frac{33(\eta_0 - \eta_2) + 48(\eta_1 - \eta_3)}{145}, \quad y = \frac{-106\eta_0 + 4\eta_1 - 39\eta_2 - 4\eta_3}{145},$$

that is, $h = 33m_0 - 9m_1 - 33m_2 - 24m_3 - 31m_4 - 20m_5 - 3m_6 + 28m_7 \in \widetilde{M}$. Since $\bar{x} = -x$, the reduced trace of h is 0, and its reduced norm $x\bar{x} + y\bar{y}$ is 5, so that $h^2 = -5$. Replacing T by another of the 6 isometries replaces h by $u^{-1}hu$ for a unit u of $\widetilde{M}$, which does not change the isometry class of the gluing (Proposition 3.3(b)). The element h is not central; it differs from the central element $\left(7 + \sqrt{29}\right)/2$ of norm 5 with which Nebe forms the modular lattices $\left(\left(7 + \sqrt{29}\right)/2\right)L_{56,2}(M)$, none of which is extremal [Neb98, Rem. 5.2]. The lattice $L_{56,2}\left(\widetilde{M}\right)$ is the $\mathrm{SL}_2(29)$-invariant unimodular lattice among the lattices $(1-\zeta_{29})^i\,\mathbb{Z}[\zeta_{29}]\widetilde{M}$ ($i \geq 0$; spanned by the products of elements of $\mathbb{Z}[\zeta_{29}]$ on the left and of $\widetilde{M}$ on the right) with the forms $\mathrm{Tr}_{\mathbb{Q}(\zeta_{29})/\mathbb{Q}}\left(c\left(x\bar{x'} + y\bar{y'}\right)\right)$ on elements $x + yb$, $x' + y'b$, where $c \in \mathbb{Q}\left(\sqrt{29}\right)$ is totally positive [Neb98, Thm. 2.4, Prop. 3.2]. Together with (1.2), these data determine L_{112} up to isometry without the files.

8. The automorphism group

We first compute, by finite exact lattice computations, the normalizer of $\mathrm{SL}_2(29)$ in $\mathrm{Aut}(L_{112})$; then we prove that $\mathrm{SL}_2(29)$ is normal in $\mathrm{Aut}(L_{112})$ by a group-theoretic argument (Appendix F). This strategy is Nebe's: she determined in this way the automorphism groups of the extremal lattices P_{48p}, P_{48q}, P_{48n} of dimension 48 and Γ_{72} of dimension 72 [Neb13, §§3.2, 6], computing the normalizer first as in [Neb98, proof of Theorem 5.3] and [Neb13, Lemma 6.1]; Sakamoto adapted it to dimension 88 [Sak26, App. A]. Neither step uses the extremality of L_{112}, and Theorem A does not depend on this section.

Theorem B. Let $S \cong \mathrm{SL}_2(29)$ act diagonally on L_{112}. (a) The centralizer $C_{\mathrm{Aut}(L_{112})}(S)$ has order exactly 48. (b) The outer automorphism of S is induced by an automorphism of L_{112}; hence $|N_{\mathrm{Aut}(L_{112})}(S)| = 2 \cdot 24\,360 \cdot 48/2 = 1\,169\,280$. (c) S is the layer $E(\mathrm{Aut}(L_{112}))$ (the product of the components); in particular S is normal, and $\mathbf{|Aut}(\boldsymbol{L_{112}})\mathbf{| = 1\,169\,280}$.

Inputs of the proof. Parts (a) and (b) are finite computations in exact arithmetic, described below [C24]. The formula in (b) uses only group theory. Since 29 is prime, $\mathrm{PSL}_2(29)$ has no field automorphisms, so $\mathrm{Out}(\mathrm{SL}_2(29)) \cong \mathrm{Out}(\mathrm{PSL}_2(29)) \cong C_2$ (generated by conjugation with an element of $\mathrm{GL}_2(29)$ of non-square determinant). Conjugation embeds $N_{\mathrm{Aut}(L_{112})}(S)/(S \cdot C_{\mathrm{Aut}(L_{112})}(S))$ in $\mathrm{Out}(S)$, and (b) says that the image is all of C_2. Since $S \cap C_{\mathrm{Aut}(L_{112})}(S) = Z(S) = \{\pm 1\}$, $|S \cdot C_{\mathrm{Aut}(L_{112})}(S)| = 24\,360 \cdot 48/2 = 584\,640$, and the factor 2 comes from (b). Part (c) is proved in Appendix F. That proof uses the classification of finite simple groups twice, through the tables of Hiss–Malle [HM01, HM02] and through the Schreier conjecture. It also uses two further finite computations, the group facts G-F1–G-F5 and the diagonal check [C25].

Let G_{112} be the Gram matrix (3.1) of L_{112}, and write endomorphisms of $V = L_{112} \otimes \mathbb{Q}$ as matrices in the basis of (3.1).

Algorithm for (a). Since $V|_S \cong V_0^2$ with $V_0 = L_{56} \otimes \mathbb{Q}$, an endomorphism of V commuting with S is a 2×2 block matrix $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $a, b, c, d \in M' \otimes \mathbb{Q}$, where $M' = \mathrm{End}_S(L_{56})$. As $V_0 \otimes \mathbb{C}|_S = 2\psi + 2\psi'$ (§2.4), $\mathrm{End}_S(V_0)$ has dimension $2^2 + 2^2 = 8$, so these X form a 32-dimensional space $\mathrm{End}_S(V)$. It has the $\mathbb{Q}$-basis $X_1, \ldots, X_{32}$ of block matrices with one block equal to a basis element of M' and the other blocks 0. The centralizer consists of the isometries of L_{112} in

$$\mathcal{H} = \{X \in \mathrm{End}_S(V) : L_{112}X \subseteq L_{112}\} = \mathrm{End}_S(V) \cap M_{112}(\mathbb{Z}).$$

To compute $\mathcal{H}$, write $X = \sum_k y_k X_k$ with $y \in \mathbb{Q}^{32}$. Each of the 112^2 entries of X is a linear form $r_e \cdot y$ with a rational coefficient row $r_e \in \mathbb{Q}^{32}$. So $X \in \mathcal{H}$ if and only if $r_e \cdot y \in \mathbb{Z}$ for all e, that is, if and only if y lies in the dual lattice $R^{\#} = \{y : r \cdot y \in \mathbb{Z} \text{ for all } r \in R\}$ of the $\mathbb{Z}$-span R of the rows r_e. We compute a basis of R by a Hermite normal form in exact integer arithmetic; R has rank 32, so $\mathcal{H}$ is a lattice of rank 32 with an explicit basis. Since the conditions $r_e \cdot y \in \mathbb{Z}$ are exactly the integrality conditions, this $\mathcal{H}$ is all of $\mathrm{End}_S(V) \cap M_{112}(\mathbb{Z})$, not a sublattice. On $\mathcal{H}$ the form $Q(X) = \mathrm{tr}(G_{112}^{-1} X G_{112} X^T)$ is positive definite and equals 112 on every isometry ($XG_{112}X^T = G_{112}$). So all isometries are among the finitely many $X \in \mathcal{H}$ with $Q(X) \le 112$, which we enumerate (Fincke–Pohst, PARI) and test exactly. To guarantee that the enumeration is complete, including rounding errors, the same vectors are enumerated again with the interval-arithmetic enumerator of §6.1 (a), which cannot miss vectors, and the two lists agree [C24].

Algorithm for (b). We take the outer automorphism α of S from GAP. The maps T on V_0 with $gT = T\alpha(g)$ for all $g \in S$ form an 8-dimensional space. We take a $\mathbb{Z}$-basis of the integral ones and repeat the computation of (a) with the 32 block matrices built from it. Every isometry found normalizes S and induces α [C24].

Table 10. Determination of $\mathrm{Aut}(L_{112})$.

Step	Result	Method	Time
(a) centralizer	$Q \le 112$: 24 elements up to sign, all isometries; $\lvert C_{\mathrm{Aut}(L_{112})}(S)\rvert = 48$	rank-32 lattice $\mathcal{H}$, enumeration	7 s
(b) outer automorphism	twisted intertwiners: 8-dimensional on L_{56}; 24 isometries up to sign	GAP for α, then as in (a)	11 min
diagonal case (Appendix F.4)	none of the 6 involutions $\neq \pm 1$ of $C_{\mathrm{Aut}(L_{112})}(S)$ splits S	integrality test	10 s
(c) normality	$E(\mathrm{Aut}(L_{112})) = S$	Appendix F (proof)	—

The centralizer of order 48 is the unitary automorphism group of the rank-2 M'-lattice $\Lambda = \left\{(a,b) \in (M')^2 : b \equiv a\tau \pmod{2M'}\right\}/\sqrt{2}$, with $L_{112} = L_{56} \otimes_{M'} \Lambda$. A matrix inducing the outer automorphism is supplied as `certificates/outer_automorphism_L112.txt` [C24, C25].

A second determination of (a). Supplement §S6 obtains the order 48 again, independently of the computation above. By Theorem S6.1(a) of the supplement, the centralizer is isomorphic to the unitary group $U(\Lambda_7)$ of a unimodular Hermitian lattice Λ_7 of rank 2 (the counterpart of Λ there). The order $\lvert U(\Lambda_7)\rvert = 48$ is computed with PARI's `qfauto` and recounted in integer arithmetic [C29, C31]. This is a different algorithm applied to a different object.

Structure. We take explicit generators: those of S, the 48 elements of $C_{\mathrm{Aut}(L_{112})}(S)$, and one element inducing the outer automorphism. We reduce them modulo 3; reduction modulo an odd prime is injective on a finite group of integer matrices (Minkowski: the kernel of $\mathrm{GL}_n(\mathbb{Z}) \to \mathrm{GL}_n(\mathbb{Z}/3\mathbb{Z})$ is torsion-free). GAP finds that they generate a group of order exactly 1 169 280 [C26]. This confirms that the elements found in (a) and (b) generate a group of order $\lvert N_{\mathrm{Aut}(L_{112})}(S)\rvert$. It is a consistency check of (a) and (b), not an independent determination of $\lvert\mathrm{Aut}(L_{112})\rvert$, which rests on Appendix F. Moreover $C_{\mathrm{Aut}(L_{112})}(S) \cong \mathrm{SL}_2(3) \circ C_4$ (SmallGroup(48,33)), with center C_4; $\mathrm{Aut}(L_{112})/S \cong C_2 \times S_4$; $\mathrm{Aut}(L_{112})/C_{\mathrm{Aut}(L_{112})}(S) \cong \mathrm{PGL}_2(29)$; $Z(\mathrm{Aut}(L_{112})) = \{\pm 1\}$; and the extension of the index-2 subgroup $S \cdot C_{\mathrm{Aut}(L_{112})}(S)$ does not split (the outer coset contains no involution) [C26]. These quotients agree with (a) and (b): $\mathrm{Aut}(L_{112})/C_{\mathrm{Aut}(L_{112})}(S)$ embeds in $\mathrm{Aut}(S) \cong \mathrm{PGL}_2(29)$ and has the same order,

and $\mathrm{Aut}(L_{112})/S$ has order 48 and contains $C_{\mathrm{Aut}(L_{112})}(S)/\{\pm 1\} \cong A_4 \times C_2$ with index 2. In short, $\mathrm{Aut}(L_{112}) \cong (\mathrm{SL}_2(29) \circ \mathrm{SL}_2(3) \circ C_4).2$, central products over the common -1. All generators have determinant 1, and the endomorphisms of L_{112} commuting with $\mathrm{Aut}(L_{112})$ form the ring $\mathbb{Z}\left[\sqrt{-29}\right]$ [C26]. Here $\sqrt{-29} = \sqrt{29} \cdot i$, where $\sqrt{29}$ generates the center $\mathbb{Q}\left(\sqrt{29}\right)$ of $\mathrm{End}_S(V)$ and i generates the center C_4 of $C_{\mathrm{Aut}(L_{112})}(S)$. An element inducing the outer automorphism exchanges ψ and ψ', so conjugation by it maps $\sqrt{29}$ to $-\sqrt{29}$; since it commutes with $\sqrt{-29}$, it maps i to $-i$. The element $\sqrt{-29}$ is skew-adjoint for the form of L_{112}, so L_{112} is a Hermitian $\mathbb{Z}\left[\sqrt{-29}\right]$-lattice of rank 56, and $\mathrm{Aut}(L_{112}) \subset \mathrm{U}(56) \subset \mathrm{SO}(112)$, where $\mathrm{U}(56)$ is the unitary group of this Hermitian structure on V.

Use of AI tools

This work was carried out with substantial use of AI systems, under the direction of the author, roiban1344, who chose the problem, made the decisions on the approach and on what to publish, and takes full responsibility for the content. The systems were Claude by Anthropic (models Claude Opus 5.5 and Claude Fable 5.1) and OpenAI Codex (models GPT-6 Astra and GPT-6.1 Sol), used in October 2026. Several models were used for overlapping tasks, so the contributions are described by system rather than by model. Claude designed the search, wrote the code, including the Lean formalization of §7.1, ran the computations, and wrote the first draft of the paper and supplement. In an earlier attempt in dimension 96, OpenAI Codex first set up the reduction of extremality to four collision types for lattices glued modulo 2; it was later used to review the arguments and to restructure and revise the paper and supplement. The separately written implementations referred to in §6 and Supplement §S3.3 were written in separate AI sessions (separate runs that did not share a working context), not by different people; what they share is stated there. The correctness of the results does not rest on trust in these tools: the proofs are written out in full, the deduction from the computational facts is checked by the Lean kernel (§7.1), the computational facts are checked in integer or interval arithmetic by the published programs (§6), and the data needed to repeat these checks are distributed with SHA-256 hashes (Appendix D).

References

[Asc00] M. Aschbacher, *Finite Group Theory*, 2nd ed., Cambridge Studies in Advanced Mathematics 10, Cambridge University Press, 2000.

[BN98] C. Bachoc, G. Nebe, Extremal lattices of minimum 8 related to the Mathieu group M_{22}, J. reine angew. Math. 494 (1998) 155–171, doi:10.1515/crll.1998.004.

[BQS95] C. Batut, H.-G. Quebbemann, R. Scharlau, Computations of cyclotomic lattices, Experiment. Math. 4 (1995), no. 3, 175–179, doi:10.1080/10586458.1995.10504319.

[BV01] C. Bachoc, B. Venkov, Modular forms, lattices and spherical designs, in: J. Martinet (ed.), Réseaux euclidiens, designs sphériques et formes modulaires, Monogr. Enseign. Math. 37, Geneva, 2001, 87–111.

[CN13] R. Coulangeon, G. Nebe, Dense lattices as Hermitian tensor products, in: Diophantine Methods, Lattices, and Arithmetic Theory of Quadratic Forms, Contemp. Math. 587, Amer. Math. Soc., 2013, 47–57, doi:10.1090/conm/587/11675; arXiv:1201.1832 (there titled "The unreasonable effectiveness of the tensor product").

[Cou00] R. Coulangeon, Tensor products of Hermitian lattices, Acta Arith. 92 (2000), no. 2, 115–130, doi:10.4064/aa-92-2-115-130.

[**FP85**] U. Fincke, M. Pohst, Improved methods for calculating vectors of short length in a lattice, including a complexity analysis, Math. Comp. 44 (1985) 463–471, doi:10.1090/S0025-5718-1985-0777278-8.
[**Gau26**] I. Gauvin, Réseaux idéaux pairs unimodulaires de rang 88 sur $\mathbb{Q}(\zeta_{115})$, preprint (15 October 2026) with an addendum; text, data and programs: repository, doi:10.5281/zenodo.23638084.
[**Gau26v**] I. Gauvin, *Independent verification of min(L112.02) = 10*, version 1.0 (26 October 2026), repository (commit `a78893f`); computer-assisted, not peer-reviewed.
[**GHY01**] W. T. Gan, J. P. Hanke, J.-K. Yu, On an exact mass formula of Shimura, Duke Math. J. 107 (2001) 103–133, doi:10.1215/S0012-7094-01-10716-3.
[**Gri99**] R. L. Griess Jr., Pieces of eight: semiselfdual lattices and a new foundation for the theory of Conway and Mathieu groups, Adv. Math. 148 (1999) 75–104, doi:10.1006/aima.1999.1846.
[**Gri10**] R. L. Griess Jr., Rank 72 high minimum norm lattices, J. Number Theory 130 (2010) 1512–1519, doi:10.1016/j.jnt.2010.02.007.
[**HM01**] G. Hiss, G. Malle, Low-dimensional representations of quasi-simple groups, LMS J. Comput. Math. 4 (2001) 22–63, doi:10.1112/S1461157000000796.
[**HM02**] G. Hiss, G. Malle, Corrigenda: Low-dimensional representations of quasi-simple groups, LMS J. Comput. Math. 5 (2002) 95–126, doi:10.1112/S1461157000000711.
[**Kin03**] O. King, A mass formula for unimodular lattices with no roots, Math. Comp. 72 (2003), no. 242, 839–863, doi:10.1090/S0025-5718-02-01455-2.
[**Kom09**] S. D. Kominers, Configurations of extremal even unimodular lattices, Int. J. Number Theory 5 (2009), no. 3, 457–464, doi:10.1142/S179304210900216X.
[**LS71**] J. Leech, N. J. A. Sloane, Sphere packings and error-correcting codes, Canad. J. Math. 23 (1971) 718–745, doi:10.4153/CJM-1971-081-3.
[**MOS75**] C. L. Mallows, A. M. Odlyzko, N. J. A. Sloane, Upper bounds for modular forms, lattices, and codes, J. Algebra 36 (1975) 68–76, doi:10.1016/0021-8693(75)90155-6.
[**Neb98**] G. Nebe, Some cyclo-quaternionic lattices, J. Algebra 199 (1998), no. 2, 472–498, doi:10.1006/jabr.1997.7163.
[**Neb12**] G. Nebe, An even unimodular 72-dimensional lattice of minimum 8, J. reine angew. Math. 673 (2012) 237–247, doi:10.1515/crelle.2011.175; arXiv:1008.2862.
[**Neb13**] G. Nebe, On automorphisms of extremal even unimodular lattices, Int. J. Number Theory 9 (2013), no. 8, 1933–1959, doi:10.1142/S179304211350067X; arXiv:1212.0865 (there titled "… of dimension 48"). The order of $\mathrm{Aut}(P_{48p})$ in Thm. 3.6 lacks a factor 2, as corrected on the author's list of publications.
[**Neb14**] G. Nebe, A fourth extremal even unimodular lattice of dimension 48, Discrete Math. 331 (2014) 133–136, doi:10.1016/j.disc.2014.05.011.
[**NP14**] G. Nebe, R. Parker, On extremal even unimodular 72-dimensional lattices, Math. Comp. 83 (2014), no. 287, 1489–1494, doi:10.1090/S0025-5718-2013-02744-5.
[**NS**] G. Nebe, N. J. A. Sloane, A Catalogue of Lattices, entries `L_56b`, `L_56c` (Nebe's lattices of dimension 56) and `dim88a`–`dim88f` (dimension 88), accessed 25 October 2026. Catalogue of Lattices.
[**Oze89**] M. Ozeki, Ternary code construction of even unimodular lattices, in: Théorie des nombres (Québec, 1987), de Gruyter, 1989, 772–784, doi:10.1515/9783110852790.772.
[**Que84**] H.-G. Quebbemann, A construction of integral lattices, Mathematika 31 (1984), no. 1, 137–140, doi:10.1112/S0025579300010731.
[**Sak26**] R. Sakamoto, Two extremal even unimodular lattices of rank 88, preprint, arXiv:2609.15633v2 (27 September 2026).
[**SE94**] C. P. Schnorr, M. Euchner, Lattice basis reduction: improved practical algorithms and solving subset sum problems, Math. Program. 66 (1994) 181–199, doi:10.1007/BF01581144.
[**Ser73**] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, 1973, Chapter VII.

[**Shi18**] I. Shimada, An even extremal lattice of rank 64, J. Number Theory 185 (2018) 1–15, doi:10.1016/j.jnt.2017.10.028.
[**Shi22**] I. Shimada, A note on Quebbemann's extremal lattices of rank 64, J. Théor. Nombres Bordeaux 34 (2022), no. 3, 813–826, doi:10.5802/jtnb.1229.
[**SPLAG**] J. H. Conway, N. J. A. Sloane, Sphere Packings, Lattices and Groups, 3rd ed., Springer 1999, doi:10.1007/978-1-4757-6568-7.
[**SSP99**] R. Scharlau, R. Schulze-Pillot, Extremal lattices, in: B. H. Matzat, G.-M. Greuel, G. Hiss (eds.), Algorithmic Algebra and Number Theory (Heidelberg, 1997), Springer 1999, 139–170, doi:10.1007/978-3-642-59932-3_8.
[**SW10**] D. Stehlé, M. Watkins, On the extremality of an 80-dimensional lattice, ANTS-IX, LNCS 6197 (2010) 340–356, doi:10.1007/978-3-642-14518-6_27.
[**Ven84**] B. B. Venkov, Even unimodular extremal lattices, Trudy Mat. Inst. Steklov. 165 (1984) 43–48 (in Russian); English translation: Proc. Steklov Inst. Math. 165 (1985) 47–52.
[**Voi21**] J. Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, Springer, 2021.
[**Wat12**] M. Watkins, Another 80-dimensional extremal lattice, J. Théor. Nombres Bordeaux 24 (2012), no. 1, 237–255, doi:10.5802/jtnb.795.
[**Soft**] The FPLLL development team, *fplll, a lattice reduction library*, version 5.5.0, source The PARI Group, *PARI/GP*, website The GAP Group, *GAP*, website Lean and mathlib, Lean source, mathlib source.
[**Repo**] roiban1344, *112-dimensional extremal even unimodular lattice* (data, the code that checks and produces the verification data, the verification data, the Lean formalization), public repository (tag `v2026-11-02`).

Appendix A. Glossary

Coset; coset minimum min(c) An element $c = x + 2L$ of $L/2L$, and the smallest norm of a vector in it. Even/odd according to $\langle x, x\rangle/2 \bmod 2$ (§2.2).
Gluing L_f The lattice $\left\{\frac{1}{\sqrt{2}}(x, v) : v \equiv xf\right\}$ built from two copies of L and an invertible parity-preserving f (Definition 3.1). Our L_{112} is L_f with $f = \tau \bmod 2$.
Collision of type (a, b) A pair (x, v) with x of norm a and v of norm b in the coset $x\tau + 2L$ (Definition 4.1). The inner-shell vectors of L_{112} are exactly the collisions of type $(6, 6), (6, 10), (10, 6), (8, 8)$ (Lemma 4.2).
Certificate (for a coset) A vector w_0 of a coset c with norm 14 or 18 (odd coset) or 12, 16, 20 (even coset) and $|\langle w_0, u\rangle| \le 6$ for all minimal vectors u; it guarantees $\min(c) \ge 14$ (odd) or $\min(c) \ge 12$ (even) (Corollaries 5.2, 5.3; Check 5.4).
Canonical form; K_8 can(c) is the smallest 56-bit integer in the $\mathrm{SL}_2(29)$-orbit of a coset c; K_8 is the set of canonical forms of cosets of norm-8 vectors. A coset contains a norm-8 vector iff its canonical form is in K_8 (Proposition 6.3).
Orbit representative A vector r_i or $x_i u$ selected from an S-orbit; not to be confused with the coset representative w_0 that certifies a bound (§§4.2, 5).

Appendix B. Extremal theta series

Scope. This appendix gives the sources of the numbers used in the text; all are exact rational computations. With $A = \theta_{E_8} = E_4$ and $B = \theta_{\Lambda_{24}} = E_4^3 - 720\Delta$, by [Ser73, Ch. VII], every modular form of weight $4k$ is a polynomial in A and B, and the extremal theta series in dimension $n = 8k$ is the unique one of the form $1 + O(q^{\lfloor n/24\rfloor+1})$.

Table B1. Extremal theta series in dimensions 56 and 112.

n	as a polynomial in A, B	first coefficients
56	$\frac{1}{240}(-159A^7 + 238A^4B + 161AB^2)$	$1 + 15\,590\,400q^3 + 36\,957\,286\,800q^4 + 15\,284\,192\,071\,680q^5 + \cdots$
112	$\frac{1}{37\,324\,800}(7\,883\,087A^{14} - 18\,165\,308A^{11}B - 38\,110\,758A^8B^2 + 74\,301\,892A^5B^3 + 11\,415\,887A^2B^4)$	$1 + 10\,888\,335\,360q^5 + 247\,006\,775\,232\,000q^6 + \cdots$

Appendix C. Proofs of auxiliary facts

C.1 τ is skew-adjoint and h is Hermitian

From $\tau G\tau^T = 5G$ and $\tau^2 = -5$ we get $G\tau^T = 5\tau^{-1}G = -\tau G$, i.e. $\langle x\tau, y\rangle = -\langle x, y\tau\rangle$. With $h(x,y) = \langle x, y\rangle + \langle x\tau, y\rangle/\sqrt{-5}$: $h(x\tau, y) = \langle x\tau, y\rangle + \langle x\tau^2, y\rangle/\sqrt{-5} = \langle x\tau, y\rangle + \sqrt{-5}\langle x, y\rangle = \sqrt{-5}\,h(x,y)$, so h is $\mathcal{O}$-linear in the first variable; and $h(y,x) = \langle x, y\rangle - \langle x\tau, y\rangle/\sqrt{-5} = \overline{h(x,y)}$ because $\overline{1/\sqrt{-5}} = -1/\sqrt{-5}$. Clearly $\mathrm{Re}\, h = \langle\cdot,\cdot\rangle$.

C.2 Proof of Proposition 3.4

Since $H = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ is free, every element of $H \otimes_{\mathcal{O}} P$ is uniquely $e_1 \otimes x + e_2 \otimes y$, and φ is a bijection onto L_τ (put $v = x\tau + 2y$). For the forms: $\mathrm{Re}(h_H(e_1,e_1)h_P(x,x')) = 3\langle x, x'\rangle = \frac{1}{2}(\langle x, x'\rangle + \langle x\tau, x'\tau\rangle)$; $\mathrm{Re}(h_H(e_2,e_2)h_P(y,y')) = 2\langle y, y'\rangle = \frac{1}{2}\langle 2y, 2y'\rangle$; and $\mathrm{Re}(\sqrt{-5}\,h_P(x,y')) = \mathrm{Re}(\sqrt{-5}\langle x, y'\rangle + \langle x\tau, y'\rangle) = \langle x\tau, y'\rangle = \frac{1}{2}\langle x\tau, 2y'\rangle$. These are the inner products of $\varphi(e_1 \otimes x) = (x, x\tau)/\sqrt{2}$ and $\varphi(e_2 \otimes y) = (0, 2y)/\sqrt{2}$, i.e. the blocks of (3.1).

C.3 Injectivity of the coset map on short shells

If $w \neq \pm w'$ have norm $m \in \{6, 8\}$ and lie in the same coset, the midpoint lemma gives $2m + 2\langle w, w'\rangle \geq 24$ and $2m - 2\langle w, w'\rangle \geq 24$; adding them gives $4m \geq 48$, which is false. So $L(6)/\{\pm 1\} \to L/2L$ and $L(8)/\{\pm 1\} \to L/2L$ are injective, with disjoint images (odd and even cosets respectively). Consequently an S-orbit of norm-8 vectors and its image in $L/2L$ have sizes in ratio $2:1$ ($-1 \in S$ acts trivially on $L/2L$), and since different canonical forms mean different orbits, $|K_8|$ equals the number of S-orbits.

C.4 Reduction of $(10, 6)$ to $(6, 10)$

If $v \equiv x\tau$ then $v\tau \equiv x\tau^2 = -5x \equiv x \pmod{2L}$. So (x, v) is a collision of type $(10, 6)$ iff (v, x) is one of type $(6, 10)$.

C.5 Orbit decomposition of $L_{56}(6)$

$632 \cdot 24\,360 + 24 \cdot 8\,120 = 15\,395\,520 + 194\,880 = 15\,590\,400$, and this is the only decomposition into 656 orbits with sizes in $\{8\,120, 24\,360\}$ summing to $|L_{56}(6)|$.

Appendix D. Data, formats and reproduction

Table D1. Data files (as distributed in [Repo], `data/`).

file	content	SHA-256 (prefix)
`L56_gram.txt`	Gram matrix G of L_{56} (56×56)	`4d13bd51…`
`sl2_29_generators.txt`	g_1, g_2 (orders 5, 7) generating $\mathrm{SL}_2(29)$	`97f88443…`
`tau.txt`	τ (56×56), $\tau^2 = -5$	`30b6fa54…`
`L112_gram_LLL.txt`	LLL-reduced Gram matrix of L_{112} (112×112)	`104764e3…`

Full SHA-256 hashes:

```
4d13bd51fcf3cb8c5706bde45828713d7eeb4d18a37e8e9d84fb2700f4bef2f0
97f88443de1fd1fdeb010ecb668dbbb03a8e0e989a18c0ad85a751687012ad90
30b6fa54f7abc2938a8f5ce29237bb92a4224f5418d3cae85183c0a89803a0f1
104764e3c8add14d124c9e4ffd011e35d4d4659f4b6216651d278d76003bbfce
```

Conventions. Row vectors; a matrix g acts by $x \mapsto xg$; isometries satisfy $gGg^T = G$; $\tau G \tau^T = 5G$. The file `L112_gram_LLL.txt` is PARI's `qflllgram` reduction of (3.1). A unimodular integer matrix U with $U \cdot (3.1) \cdot U^T$ equal to this file is supplied as `certificates/reduction_L112.txt` and checked in integer arithmetic [C2], so both files describe the same lattice.

Specification without the files. The defining matrices (56×56 and 112×112) are not printed in this paper; they are the files of Table D1. By §7.2, L_{56} is $\mathrm{SL}_2(29)$-equivariantly isometric to $L_{56,2}\big(\widetilde{M}\big)$ of [Neb98], and in those coordinates τ is right multiplication by an element of $\widetilde{M}$ of reduced norm 5 and reduced trace 0, given in §7.2. Together with (1.2), this specifies L_{112} up to isometry independently of the data files. The files remain the reference: every verification program reads them and checks their hashes.

Verification data (orbit representatives of $L_{56}(6)$ and $L_{56}(8)$, odd and even-coset certificates, basis changes, witnesses of norm 8 for B and the data of the supplementary computations) are distributed in `certificates/` of [Repo] and its Release `certificates-v2`. The file `certificates/SHA256SUMS` lists the SHA-256 hash of each of them, and every program checks a file against it before use. The first line below is the SHA-256 hash of `SHA256SUMS` itself, which thus fixes all verification data; the other lines are the hashes of the files read by `code/verify/proof.sh` (and of `even_bounding.npz`, read by `full.sh`).

SHA-256:

```
70edb97353eb99c5ecbef64a48c4210963b1c76c1b58d50bebdc555631bf6293  SHA256SUMS
db60e9fa296f4b394ee728b21ab8a67a2a6d3289b6f61f80e1c3455e319ca261  reduction_L112.txt
5ed0cc64aa6c13fcb2402c565b367b7bf43c9e085765b91881d1fb07a6d2dd59  reduction_L.txt
a74975e31bbe96952c48cf3eb9239f8acbe86635a358f834cd2141f0963aca74  min_reps.npz
e8bbdfaae0293cd0d4609d19ec284c527ab6a5705fedc4661339c4c21bf780ef  centralizer.npz
c451b3b6b0935d69e0fe62cc3250bfd0c7642ceba8403d56e933aa22584a9f3a  norm8_reps.npz
8008c8c0d9ccbf5299d87ba03b06c1e176b5441ec9fe542a2c9d6402ccf12915  odd_bounding.npz
56f01969dc7ec444ede584082bab6c46ccec287007968c3acc8af9478ef0008e  even_bounding.npz
```

Only the four data files of Table D1 must be the right ones for Theorem A: the programs check again every property of the verification data that the proof uses, so altered verification data can make a check fail but not pass; the hashes record which files were checked.

Elementary checks (a few seconds in any computer algebra system): (3.1) has even diagonal and determinant 1; $\tau^2 = -5$, $\tau G \tau^T = 5G$; $g_i G g_i^T = G$, $g_i \tau = \tau g_i$. Checking one odd-coset certificate, given F1 and F2: (a) $w_0 - r\tau$ has even coordinates, (b) $\langle w_0, w_0 \rangle \in \{14, 18\}$, (c) $|\langle w_0, r' g \rangle| \leq 6$ for all 656 representatives r' and all 12 180 elements g.

Code. Every computation used in the proof of Theorem A, and every supplementary computation cited with a label, is listed in Table D2 with the code that performs it, at its path in the public repository [Repo]. The history of the search in Supplement §S5 is given as motivation only; its code is not distributed. The text refers to the rows of Table D2 as [C1], [C2], ….

Checking the computational facts from the supplied data takes about 1.2 CPU hours by the fastest route (route A of Supplement Table S7; the script `code/verify/proof.sh`). This uses the distributed basis change for F1; computing the BKZ-40 basis again (`code/produce/p6_reduction.py`) takes about 1 minute more. Alternative routes and additional tests are described in Supplement §S3.

Table D2. Computations and code. References such as [C1] in the text point to the rows of this table. **Bold**: the checks run by `code/verify/proof.sh` (route A of Supplement Table S7), which together establish F0–F5. The directory gives the role of a file: `verify/` checks (v7 is the alternative check of F5b, v8 a consistency test), `produce/` produced the verification data (not needed for the proof), `crosscheck/` is implementation 2 (separately written code), `supplementary/` contains computations not used in the proof of Theorem A, `lean/` the Lean formalization. Times are elapsed times of single-threaded runs unless stated otherwise, as given in the docstring of each file.

ID	computation and code (public-repository paths unless marked otherwise)	time
C1	Extremal theta series of L and L_{112}; Siegel means $\Sigma(n)$; Siegel orbit excess (exact rational arithmetic) **code/verify/v2_theta.py**	a few s
C2	Basic facts F0, F4 (`v0_inputs.py`): (3.1) is even with determinant 1; g_1, g_2 preserve G, commute with τ and generate a group of order 24 360 containing -1; $\tau^2 = -5$, $\tau G \tau^T = 5G$. Not used in the proof: the group is isomorphic to $\mathrm{SL}_2(29)$ (GAP, `gluings.py` (e)); the centralizer has order 12 (`p1_centralizer.py`); the 24 elements with $\tau^2 = -5$ form 2 classes (`gluings.py` (d)) **code/verify/v0_inputs.py** code/supplementary/gluings.py code/produce/p1_centralizer.py	v0 20 s gluings 1 min p1 10 min
C3	Gluings: the commutant of $\mathrm{SL}_2(29)$ on $L/2L$ is 8-dimensional (the image of M'); 60 invertible parity-preserving maps; their classes (Proposition 3.3, Table 2); forced collisions code/supplementary/gluings.py code/supplementary/nebe/gluings_m.py code/supplementary/_shared.py	gluings 1 min gluings_m 30 min
C4	Gluings A and B: minimum 8 and exactly 292 320 vectors of norm 8; invariants of A and B compared code/supplementary/gluing_b.py code/supplementary/nebe/gluings_m.py code/supplementary/_shared.py	gluing_b 15 min gluings_m 30 min
C5	F1: $\min L = 6$ by Fincke–Pohst enumeration with exact rational LDL^T and outward-rounded interval arithmetic (BKZ-40 basis, 8 454 837 756 nodes); controls E_8, Leech, Leech $\perp$ Leech $\perp E_8$, E_8 in 5 skewed bases, and 60 random positive definite forms compared with exhaustive search **code/verify/v1_min_L.py** **code/verify/enum_interval.c** code/produce/p6_reduction.py	v1 34 min p6 1 min
C6	F1 by reduction to $L_{56,2}(M)$: no class of a norm-8 vector of $L_{56,2}(M)$ lies in the invariant subspace W (linear algebra over $\mathbb{F}_2$); positive control code/supplementary/nebe/reduction_to_nebe.py code/supplementary/nebe/lattices.py code/supplementary/nebe/tables.py	30 min (tables 3 h)

ID	computation and code (public-repository paths unless marked otherwise)	time
C7	F1 by floating-point enumeration (fplll on two bases, PARI `qfminim`); supplementary `code/supplementary/float_enumerations.py`	svp 2 × 10–30 min qfminim 30 min

C8	F2: the 656 orbit representatives of $L(6)$; distinct orbits whose sizes add up to 15 590 400; spherical design identities of degree 2, 4, 6 (consistency test, `v8`); the expansion, disjointness and the identities for $m = 6$ by implementation 2 (`minimal_orbits.py`)
	`code/produce/p2_minimal_orbits.py`
	`code/verify/v3_minimal_orbits.py`
	`code/verify/v8_design_identities.py`
	`code/crosscheck/second_impl/minimal_orbits.py`
		p2 9 min
		v3 3 min
		v8 14 min
		minimal_orbits 80 s

C9	F3: orbit representatives of $L(8)$ by marking sums of two minimal vectors; 1 517 130 distinct orbits whose sizes add up to 36 957 286 800; design identities (`v8` on the output of `v4`; `design_identities.py` of implementation 2 on the S^*-orbits of `norm8_reps.npz`)
	`code/produce/p3_norm8_orbits.py`
	`code/verify/v4_norm8_orbits.py`
	`code/verify/v8_design_identities.py`
	`code/crosscheck/second_impl/design_identities.py`
		p3 1.2 h
		v4 9 min
		design_identities 18 min

C10	F5a: odd-class certificates for the 656 classes $r\tau + 2L$ (norms 14, 18; inner products with all minimal vectors at most 6), and their re-check by separately written code
	`code/produce/p4_odd_descent.py`
	`code/verify/v5_odd_bounding.py`
	`code/crosscheck/second_impl/odd_classes.py`
		p4 6 min
		v5 2 min
		odd_classes 1.5 min

C11	F5b by canonical forms on $L/2L$: the table K_8 and the look-up for all 1 527 984 pairs, in two implementations (`v6`; `even_classes.py`, also for gluing B); implementation 2′s sample of 180 pairs by enumerating $\langle 2L, t\rangle$ with PARI, with the 10 colliding pairs of B as positive control (`coset_enumeration.py`; not a proof)
	`code/verify/v6_canonical_lookup.py`
	`code/crosscheck/second_impl/even_classes.py`
	`code/crosscheck/second_impl/coset_enumeration.py`
		v6 24 min
		even_classes 1.7 h
		coset_enumeration 1.7 h

C12	Closest-vector enumerations (fplll, floating point) for the classes of F5a and F5b; supplementary `code/supplementary/float_enumerations.py`	cvp610 2.5 h cvp88 4 × 9 h

C13	F5b by even-class certificates for all 1 527 984 pairs (norms 12, 16, 20; inner products with all minimal vectors at most 6), and their check
	`code/produce/p5_even_descent.py`
	`code/verify/v7_even_bounding.py`
		p5 6 × 31 h
		(190 CPU h)
		v7 24 CPU h
		(7.9 h, 3 threads)

C14	Automorphisms preserving the standard frame; the halving of the pairs in route B `code/supplementary/frame_stabilizer.py`	seconds

ID	computation and code (public-repository paths unless marked otherwise)	time
C15	Route S: random sample of certificates re-checked with independent integer code `code/supplementary/sample_check.py`	10 min
C16	Gaussian estimate of the number of enumeration nodes for L_{112} itself (LLL, BKZ-20, BKZ-30 bases) `code/supplementary/node_estimate.py`	a few s
C17	Frame-change test: all 864 frames of Λ give the same class `code/supplementary/frames.py`	7 min
C18	Fixed sublattices of elements of order 3, 5, 7 and their exact minima; BKZ on L_{112} (tests that cannot separate B from C) `code/supplementary/fixed_sublattices.py`	L_{112} 10 min B 18 min
C19	Support modules of the norm-8 vectors of B (Supplement Table S9) `code/supplementary/anatomy_b.py`	a few s

C20	Supplement §S4.1, cross-check: rank-2 support modules (free and non-free), grouped into 97 invariant types, 63 dangerous (first enumeration with $a \leq 32$). Both enumerations agree on 58 dangerous types after regrouping by the same coarser invariant
	`code/supplementary/route_t/classify.py`
	`code/supplementary/route_t/recount.py`
	`code/supplementary/route_t/hermitian.py`
		1–2 min each

C21	Supplement §S4.1, cross-check (not needed for route T): three searches modulo 2. All 3 936 certificates for test (3) checked in integers. Test (2): all 763 992 representatives generated, none unresolved; all checked in integers (class membership, norm 12, 16 or 20, inner-product bound 6 against every minimal vector). Complete data: `certificates/route_t_search2.npz`
	`code/supplementary/route_t/certify.py`
	`code/supplementary/route_t/check.py`
	`code/supplementary/route_t/float_cvp.py`
	`code/supplementary/route_t/common.py`
		(2): certify
		6 CPU days
		check 6 CPU h
		(5.0 h, 2 threads)
		(3): certify 25 min
		check 2 min

C22	Lean formalization of the deduction from F1–F5 to Theorem A (F5b by certificates or by canonical forms), and its check: build without warnings, axioms as in `expected-axioms.txt` (standard only), kernel replay by `leanchecker`
	`lean/EEUL/L112/Chain.lean`
	`lean/CheckAxioms.lean`
	`lean/check.sh`
	`lean/expected-axioms.txt`
		build 1 min
		–fresh 3 min

C23	Identification of L with Nebe's $L_{56,2}(\widetilde{M})$ built from the definition; τ as right multiplication by an element of reduced norm 5 `code/supplementary/nebe/identify_nebe.py`	45 min

ID	computation and code (public-repository paths unless marked otherwise)	time
	code/supplementary/nebe/lattices.py code/supplementary/nebe/cycloquat.py	
C24	The centralizer of $\mathrm{SL}_2(29)$ in $\mathrm{Aut}(L_{112})$ (order 48) and an automorphism of L_{112} inducing the outer automorphism code/supplementary/automorphisms/aut_normalizer.py code/supplementary/automorphisms/aut_common.py code/supplementary/_shared.py	15 min
C25	Finite group facts G-F1–G-F5 used in Appendix F (GAP), and the diagonal case; the simple groups in the rows of degree ≤ 112 of the Hiss–Malle Table 3 with their orders (Table F1), and further checks G-F6–G-F12 code/supplementary/automorphisms/group_facts.g code/supplementary/automorphisms/out_facts.g code/supplementary/automorphisms/aut_diagonal.py code/supplementary/automorphisms/aut_common.py code/supplementary/_shared.py	GAP 10 s each aut_diagonal 1 min
C26	Structure of $\mathrm{Aut}(L_{112})$: order of the generated group, non-split extension, quotients, Molien series; the $\mathbb{Z}\left[\sqrt{-29}\right]$- and $\mathcal{O}_K$-structures code/supplementary/automorphisms/aut_structure.py code/supplementary/automorphisms/aut_character.py code/supplementary/automorphisms/endomorphisms.py code/supplementary/automorphisms/aut_common.py code/supplementary/_shared.py	aut_structure 30 min aut_character 40 min endomorphisms 3 min
C27	Independent re-check of all certificates and of F1 by separately written code (integer arithmetic; float32 products only with proved bounds on all partial sums) code/crosscheck/all_data/check_all.py code/crosscheck/all_data/min_L.py code/crosscheck/all_data/min_L_enum.c	check_all 10 h min_L 96 min
C28	Gluings A and B are isometric: an explicit basis change T with $TG_BT^\top = G_A$, checked in integer arithmetic ($\det T = \pm 1$) code/supplementary/isometry_ab.py code/supplementary/_shared.py	find 8 min check a few s
C29	The $\mathrm{SL}_2(29)$-invariant even unimodular lattices of dimension 112 (`local`, `mass`, `enum`): the image of the group ring is maximal at $p = 2, 3, 5, 7, 29$ ($\mathbb{F}_p$-algebras of dimension 392), so every such lattice is $L_{56,2}(M) \otimes_M \Lambda$; $\zeta_K(-1) = 1/2$, $\zeta_K(-3) = 157/20$ ($K = \mathbb{Q}\left(\sqrt{29}\right)$, generalized Bernoulli numbers) and the mass $157/640$; Kneser neighbours at the prime of norm 5 (fixed random seed) reach the mass with 8 classes, $\lvert U(\Lambda)\rvert = 1152, 12, 288, 24, 24, 20, 288, 48$ (PARI `qfauto`) code/supplementary/genus/genus.py code/supplementary/genus/genus_steps.py code/supplementary/genus/genus_lattice.py code/supplementary/genus/genus_data.py	local 20 min; mass 1 s; enum 16 min

ID	computation and code (public-repository paths unless marked otherwise)	time
C30	The 8 lattices $X_j = L_{56,2}(\mathfrak{M}) \otimes_{\mathfrak{M}} \Lambda_j$ (`build`; Gram matrices and bases in the coordinates of (3.1) in the verification data), their mod-2 frames (`frames`), and vectors of norm 8 in classes 3, 4, 5 from fixed sublattices (`witness`) code/supplementary/genus/genus.py code/supplementary/genus/genus_steps.py code/supplementary/genus/genus_lattice.py code/supplementary/genus/genus_data.py	build 4 min; frames 12 min; witness 2 min
C31	Explicit $\mathrm{SL}_2(29)$-equivariant isometries class 7 $\to L_{112}$ and class 3 $\to$ gluing A, and a recount of $\lvert U(\Lambda_j)\rvert$ as equivariant automorphisms (`identify`); integer-arithmetic check, numpy only (`check`): each G_X even with determinant 1, g_1, g_2 act on X_j, the vectors of norm 8, the isometries ($\det T = \pm 1$), and $\sum_j 1/\lvert U(\Lambda_j)\rvert = 157/640$ code/supplementary/genus/genus.py code/supplementary/genus/genus_steps.py code/supplementary/genus/genus_lattice.py code/supplementary/genus/genus_check.py code/supplementary/genus/genus_data.py code/supplementary/_shared.py	identify 13 min; check 80 s
C32	Supplement §S4.1, Proposition S4.3 (route T, final step): the range of Proposition S4.2 enumerated by the checker itself (63 996 parameter tuples, 1 171 valid modules, 631 dangerous); for each dangerous module a witness contradicting F5b (541) or F5a (90), checked in integers by an enumeration independent of the producer (`check`, Python standard library only); `make` reproduces the witnesses code/supplementary/route_t/exclusion.py certificates/route_t_exclusion.json	check 8 min make 1 min
C33	Nebe's tensor products [Neb98, §5] for $(p,q) = (5,29)$, not used for Theorem A: M contains the Hurwitz order and $\widetilde{M}$ a maximal order of $\mathcal{Q}_{\infty,3}$ (24 and 12 units of reduced norm 1); for every pair of classes of embeddings $\Lambda = L_{8,2} \otimes_{\mathcal{O}} L_{56,2}$ has rank 112 and $\Lambda^{\#} = \Lambda\mathfrak{a}$ with $\mathfrak{a}$ not principal, while $\mathbb{Q}\big(\sqrt{5},\sqrt{29}\big)$ has class group $\mathbb{Z}/2$ (narrow as well), so no invariant form, even on $\Lambda\mathfrak{b}$, is unimodular; control $(5,13)$: even unimodular, no vector of norm ≤ 4 (L_{48}). Class numbers 1 for $\mathbb{Q}\big(\sqrt{5},\sqrt{13}\big)$, $\mathbb{Q}\big(\sqrt{5},\sqrt{17}\big)$ code/supplementary/nebe/tensor_neb98.py code/supplementary/nebe/cycloquat.py	20–36 min
C34	Sampled checks of the table K_8 and of the canonical-form test (not needed for the proof): positive control on 3 000 pairs for a gluing along an element of C (Table 6); the classes of 2 000 norm-8 vectors $a \mp b$ (a, b minimal) are in K_8 (§6.4); agreement with closest-vector enumeration (fplll, radius^2 2.05) on 150 random pairs for τ, 150 for gluing B and all colliding pairs of gluings B (`k8_checks.py`)	k8_checks 3 min

ID	computation and code (public-repository paths unless marked otherwise)	time
	and A (`gluings_m.py` (e)) code/supplementary/k8_checks.py code/supplementary/nebe/gluings_m.py	

Appendix E. Pseudocode

```
MINIMAL-ORBITS(L, S):
  candidates <- initial minimal vectors; reps <- []; seen <- {}; total <- 0
  while total < 15590400:
    r <- a candidate outside seen; O <- {r*g : g in S}
    append r to reps; seen <- seen union O; total <- total + size(O)
    for s in O with <r,s> = 3: add r-s to candidates
    if no unused candidates remain: enlarge the initial candidate list
  check distinct orbits and total = 15590400
  return reps

NORM8-ORBITS(L, S_star, minimal_reps):
  for r_i in minimal_reps and s in L(6) with <r_i,s> = -2:
    x <- r_i+s
    if x is marked: continue
    find and mark all marked elements of x*S_star using <x*u,r_j*g> = +/-4
    record x and size(S_star)/size(stabilizer(x))
  check distinct orbits and total orbit size = 36957286800

CANONICAL(c, S):
  return min { pack56(c*g mod 2) : g in S/{+1,-1} }

TEST88(pairs, K8, tau):
  for (x_i,u) in pairs:
    if CANONICAL(x_i*u*tau, S) in K8: report collision

COSET-REPRESENTATIVE(w):
  repeat:
    m <- max { <w,u> : u in L(6) }
    if m <= 6: return w
    choose u with <w,u> = m; w <- w-2*u
  # Check 5.4 also verifies parity and the allowed final norm.
```

Appendix F. Proof that $\mathrm{SL}_2(29)$ is normal in $\mathrm{Aut}(L_{112})$

Write $A = \mathrm{Aut}(L_{112})$, $S \cong \mathrm{SL}_2(29)$, $V = L_{112} \otimes \mathbb{Q}$, so $V|_S \cong V_0^2$ with $V_0 = L_{56} \otimes \mathbb{Q}$ irreducible of dimension 56 and $V \otimes \mathbb{C}|_S = 4\psi + 4\psi'$, where ψ, ψ' are the two faithful characters of degree 14. The generalized Fitting subgroup $F^*(A) = F(A)E(A)$ satisfies $C_A(F^*(A)) \le F^*(A)$ (Bender). We show (F.1) that every normal subgroup of A has at most two isotypic components on V; (F.2, F.3) that $F(A)$ centralizes S; and (F.4) that $E(A) = S$. The architecture of the proof is that of Nebe [Neb13, §3.2, §6], as adapted by Sakamoto [Sak26, App. A]. Both use the primitivity of the action and Ph. Hall's theorem on the p-groups all of whose characteristic abelian subgroups are cyclic. Primitivity fails here (there are two isotypic components a priori); F.1 replaces it, and F.3 replaces Hall's theorem by a character-degree argument. The finite facts G-F1–G-F5 below were checked with GAP [C25]. G-F1–G-F3 and G-F5 are

computed from the groups themselves; G-F4 uses the character tables of Ru, 2. Ru and $2.L_2(29)$ and the list of maximal subgroups of Ru in GAP's character table library CTblLib.

Finite facts. G-F1: the proper subgroups of $\mathrm{PSL}_2(29)$ of index ≤ 112 are $29:14$ (index 30) and $29:7$ (index 60). G-F2: every proper subgroup of $\mathrm{SL}_2(29)$ has index ≥ 30. G-F3: the irreducible characters of the preimages in S of $29:14$, $29:7$ have degrees $1, 14$ and $1, 7$; ψ, ψ' restricted to a subgroup U of order 29 do not contain the trivial character. G-F4: the only maximal subgroup of Ru of order divisible by 29 is $L_2(29)$; both characters of degree 28 of 2. Ru restrict irreducibly to $2.L_2(29)$. The restriction depends on the class fusion of $2.L_2(29)$ into 2. Ru; GAP finds 6 maps of conjugacy classes compatible with the two character tables, the actual fusion is among them, and the claim holds for all 6. G-F5: $|\mathrm{SL}_2(29)|$ divides neither $|\mathrm{SL}_2(59)|$ nor $|\mathrm{SL}_2(173)|$.

F.1 At most two isotypic components

Let $N \trianglelefteq A$ and let $V_1, ..., V_b$ be the isotypic components of $V|_N$ over $\mathbb{Q}$. A permutes them (Clifford), hence so does S, through $\mathrm{PSL}_2(29)$ since -1 is scalar. If the permutation action were nontrivial, a nontrivial orbit would have a stabilizer of index ≤ 112, so of index 30 or 60 (G-F1), and its components would have dimension ≤ 3. The sum of the orbit would be induced from a representation θ of dimension ≤ 3 of the preimage J of the stabilizer; by G-F3, θ is a sum of linear characters, trivial on $U \leq [J, J]$. A common constituent of θ and $\psi|_J$ would give a trivial constituent of $\psi|_U$, contradicting G-F3; so by Frobenius reciprocity $\mathrm{Ind}_J^S \theta$ contains neither ψ nor ψ', which is impossible inside $V|_S$. Hence S fixes every V_i, each V_i is an S-submodule of V_0^2 of dimension 56 or 112, and $\boldsymbol{b \leq 2}$.

F.2 Odd primes

Let $P_0 \trianglelefteq A$ be a p-group, p odd, let V_1 (and V_2) be the isotypic components of $V|_{P_0}$ (F.1), and let K_j be the kernel on V_j, so $K_1 \cap K_2 = 1$. Consider j with $P_0/K_j \neq 1$. Then P_0/K_j has a faithful irreducible rational representation U_j with $V_j \cong U_j^m$. For a constituent χ of $U_j \otimes \mathbb{C}$, $\dim U_j = [\mathbb{Q}(\chi) : \mathbb{Q}]\chi(1)$, since the Schur index is 1 (Roquette, p odd). The center $Z(P_0/K_j) \neq 1$ acts by scalars through a faithful linear character λ, so it is cyclic of order p^f ($f \geq 1$), and $\chi(z) = \chi(1)\lambda(z)$. Hence $\mathbb{Q}(\chi)$ contains $\mathbb{Q}(\zeta_{p^f})$; it is contained in a cyclotomic field $\mathbb{Q}(\zeta_{p^k})$, because character values of a p-group are sums of p-power roots of unity. So $[\mathbb{Q}(\chi) : \mathbb{Q}] = (p-1)p^e$ for some $e \geq 0$, and $\dim U_j = (p-1)p^e\chi(1)$ divides $\dim V_j \in \{56, 112\}$. As $112 = 2^4 \cdot 7$, this forces $p^e\chi(1) = 1$ and $p - 1 \mid 112$, i.e. $p \in \{3, 5, 17, 29, 113\}$ (the prime 7 divides 112, but $7 - 1 = 6$ does not), and then $P_0/K_j \cong C_p$. Thus $P_0 \hookrightarrow C_p \times C_p$, and the image of S in $\mathrm{Aut}(P_0) \subseteq \mathrm{GL}_2(p)$ is a quotient of $\mathrm{SL}_2(29)$, so it is trivial unless 29 divides $|\mathrm{GL}_2(p)| = p(p-1)^2(p+1)$. For $p \neq 29$, $29 \nmid |\mathrm{GL}_2(p)|$, so S acts trivially on P_0. For $p = 29$: if P_0 is cyclic, $\mathrm{Aut}(P_0)$ is abelian; if $P_0 \cong C_{29}^2$ then $b = 2$ and S fixes K_1 and K_2 (F.1), so its image in $\mathrm{GL}_2(29)$ lies in a split torus, which is abelian. In all cases $[P_0, S] = 1$.

F.3 The prime 2

Let $P = O_2(A)$. S is generated by its elements of order 29, so it suffices to show that such an element y centralizes P. Let D_0 be a critical subgroup of P in the sense of Thompson: D_0 is characteristic in P, of class ≤ 2, and every nontrivial automorphism of P of odd order acts nontrivially on D_0. Since $D_0 \trianglelefteq A$, F.1 applies: $V|_{D_0}$ has isotypic components V_1 (and V_2), fixed by y; let K_j be the kernel on V_j, so $K_1 \cap K_2 = 1$ and y acts on $Q_j = D_0/K_j$. A constituent χ of $V_j \otimes \mathbb{C}$ is a faithful irreducible character of Q_j (Galois conjugates have the same kernel) with $\chi(1) \leq 112$.

Lemma. If Q has class ≤ 2 and $\chi \in \mathrm{Irr}(Q)$ is faithful, then $\chi(1)^2 = |Q : Z(Q)|$. *Proof.* For $g \notin Z(Q)$ pick h with $z = [g,h] \neq 1$; $z \in Z(Q)$ acts as a scalar $\lambda(z) \neq 1$, and $\chi(g) = \chi(h^{-1}gh) = \chi(gz) = \lambda(z)\chi(g)$, so $\chi(g) = 0$. The orthogonality relation $\sum_g |\chi(g)|^2 = |Q|$ gives $\chi(1)^2 |Z(Q)| = |Q|$. ■

Hence $|Q_j : Z(Q_j)| \le 112^2 < 2^{14}$, and $Z(Q_j)$ is cyclic. The element y acts trivially on $Z(Q_j)$ (the automorphism group of a cyclic 2-group is a 2-group) and on the abelian group $Q_j/Z(Q_j)$: otherwise it would act nontrivially on its Frattini quotient (Burnside), an $\mathbb{F}_2$-space of dimension ≤ 13, whereas nontrivial irreducible $\mathbb{F}_2[C_{29}]$-modules have dimension 28 (the order of 2 modulo 29). Writing $x^y = x\,\zeta(x)$ with $\zeta : Q_j \to Z(Q_j)$ a homomorphism, $x = x^{y^{29}} = x\,\zeta(x)^{29}$ forces $\zeta = 1$. So $[D_0, y] \le K_1 \cap K_2 = 1$, and by the property of D_0, y centralizes P.

F.4 The layer

By F.2–F.3, $F(A) \le C_A(S)$. Let $E(A) = Q_1 \cdots Q_r$ be the product of the components (commuting quasisimple subnormal subgroups).

(a) *S normalizes every Q_i.* By F.1 for $E(A)$, $V|_{E(A)}$ has at most two isotypic components; the complex constituents of the j-th are Galois conjugates of $X_j = \bigotimes_i X_{j,i}$. Each Q_i acts nontrivially, nontrivial irreducible representations of quasisimple groups have degree ≥ 2, and $2^7 > 112$, so at most 6 of the Q_i act nontrivially on each of the at most two components; hence $r \le 12$. By G-F2, S cannot permute the Q_i nontrivially.

(b) *S induces inner automorphisms on each Q_i.* By the Schreier conjecture, a consequence of the classification theorem, the outer automorphism group of the simple group $\frac{Q_i}{Z(Q_i)}$ is solvable. Automorphisms of a quasisimple group act faithfully on its central quotient, so $\mathrm{Out}(Q_i)$ is solvable as well. Since S is perfect, its image in $\mathrm{Out}(Q_i)$ is trivial. Thus $S \le E(A)C_{A(E(A))}$.

(b′) $S \le E(A)$. Put $T = SE(A) \cap C_A(E(A))$; then $SE(A) = E(A)T$ and $T/Z(E(A)) \cong SE(A)/E(A)$ is perfect. As S and $E(A)$ centralize $F(A)$, $T \le C_A(F^*(A)) \le Z(F^*(A))$ is abelian. So $T \le E(A)$ and $S \le E(A)$.

(c) *S lies in a single component.* Let I be the set of i for which the projection of S to $Q_i/Z(Q_i)$ is nontrivial. $I \neq \emptyset$, since otherwise S would lie in the abelian group $Z(E(A))$. For $i \in I$, let $H_i \le Q_i$ be the preimage of the projection of S, a nontrivial quotient of $\mathrm{SL}_2(29)$, and let $R_i = [H_i, H_i]$. Then R_i is a perfect central extension of $\mathrm{PSL}_2(29)$, hence a quotient of its universal cover $\mathrm{SL}_2(29)$, so its nontrivial irreducible representations have degree $\ge (29-1)/2 = 14$. A nontrivial irreducible representation of Q_i is nontrivial on R_i: its kernel is a proper normal subgroup of the quasisimple group Q_i, hence central, while R_i is not central. So the nontrivial irreducible representations of Q_i have degree ≥ 14. As $14^2 > 112$, each X_j involves at most one $i \in I$, so $|I| \le 2$.

Suppose $|I| = 2$, say $I = \{1,2\}$. Then $b = 2$, and after renumbering Q_1 acts trivially on V_2 and Q_2 trivially on V_1. Write $s \in S$ as $s = zq_1q_2$ ($z \in Z(E(A))$, $q_i \in Q_i$), and similarly $t = z'q_1'q_2'$. Then $[s,t] = [q_1,q_1'][q_2,q_2']$, and $[q_1,q_1'] \in A$ acts as $[s,t]$ on V_1 and trivially on V_2. Let $\varphi(s)$ act as s on V_1 and as the identity on V_2; φ is a homomorphism from S to $\mathrm{GL}(V)$, and since S is perfect, it maps S into A. The element $c = \varphi(-1)$ acts as -1 on V_1 and as $+1$ on V_2; it is an involution in $C_A(S)$ other than ± 1, and $\varphi(g) = \frac{1-c}{2}g + \frac{1+c}{2}$ for $g \in S$, because $(1 \mp c)/2$ are the projections onto V_1 and V_2. As φ is a homomorphism, $\varphi(S) \subseteq A$ holds if and only if $\varphi(g_1)$ and $\varphi(g_2)$ preserve L_{112} for the two generators g_1, g_2 of S.

Lemma (diagonal check). For each of the 6 involutions $c \neq \pm 1$ of $C_A(S)$ (the group of order 48 of Theorem B(a)), the matrix $\frac{1-c}{2}g + \frac{1+c}{2}$ does not preserve L_{112} for at least one $g \in \{g_1, g_2\}$ [C25].

So $|I| = 2$ is impossible. Hence $|I| = 1$ and $S = [S, S] \le [Q_1, Q_1] = Q_1$.

(d) $E(A) = Q_1$. Any other component centralizes $Q_1 \supseteq S$ and hence lies in $C_A(S)$ of order 48; but quasisimple groups have order ≥ 60.

(e) $Q_1 = S$. Let X be an irreducible constituent of $V \otimes \mathbb{C}|_{Q_1}$; it is absolutely irreducible and in characteristic 0. Its restriction to S is contained in $V \otimes \mathbb{C}|_S = 4\psi + 4\psi'$, so it is a sum of copies of ψ and ψ' and is faithful on S. Hence $\bar{Q} = Q_1/\ker X$ is quasisimple ($\ker X$ is a proper normal, hence central, subgroup of Q_1), S embeds in $\bar{Q}$, so $29 \mid |\bar{Q}|$, and the image of $-1 \in S$ is a central involution, so $Z(\bar{Q})$ has even order. Moreover X is a faithful absolutely irreducible representation of $\bar{Q}$ of degree ≤ 112.

The groups $\bar{Q}$ allowed by Hiss–Malle. By [HM01, Theorem 1], for a quasisimple group G and a faithful absolutely irreducible representation of degree $d \le 250$, the pair (G, d) is in [HM01, Table 2] or [HM01, Table 3]. The only exclusion in that theorem, groups of Lie type in their defining characteristic, is empty in characteristic 0. The corrigenda [HM02] add the representations of $L_3(3)$ and the degree-248 representations of $L_4(5)$, and reprint the corrected Table 3 as [HM02, Table 2]; this changes nothing below, as $29 \nmid |L_3(3)|$ and $248 > 112$. We go through the entries.

- [HM01, Table 2(a)]: the natural modules of A_n. Here $G = A_n$ has trivial center.
- [HM01, Table 2(b)–(d)]: $L_2(q)$ and $2.L_2(q)$. A center of even order means $G = 2.L_2(q) = \mathrm{SL}_2(q)$ with q odd. Its faithful degrees are $(q-1)/2$ (if $q \equiv 1 \bmod 4$) or $(q+1)/2$ (if $q \equiv 3 \bmod 4$), $q-1$ and $q+1$. The smallest of them is ≤ 112, and $29 \mid q(q^2-1)$, only for $q \in \{29, 59, 173\}$.
- [HM01, Table 3], which omits the entries of Table 2 for A_n with $n \ge 14$ and for $L_2(q)$ with $q \ne 9$ [HM01, §6]. The simple groups $G/Z(G)$ occurring in its rows of degree ≤ 112 (78 groups, including $L_3(3)$ from [HM02]) are listed in `out_facts.g` [C25], and their orders are prime to 29 except for Ru and J_4. The rows concerned are Ru (degree 28, characteristic 2 only), J_4 (degree 112, characteristic 2 only) and 2. Ru (degree 28, characteristic $\ne 2$, two representations). Ru and J_4 have trivial center.
- The double covers $2.A_n$, whose faithful representations are not in Table 2. They are in Table 3 for $n \le 13$, and for $n \ge 14$ they have a faithful representation of degree ≤ 250 only if $n \le 18$ [HM01, Corollary 6]. But $29 \mid |2.A_n|$ requires $n \ge 29$. (Consistently, the basic spin representations of $2.A_n$ have degree $\ge 2^{\lfloor (n-2)/2 \rfloor}$, which is $\ge 2^{13}$ for $n \ge 29$.)

Hence the quasisimple groups with a center of even order, order divisible by 29, and a faithful irreducible character of degree $d \le 112$ in characteristic 0 are exactly those of Table F1 (the degrees and numbers of characters are those of the character tables of $\mathrm{SL}_2(q)$, checked with GAP).

Table F1. Quasisimple groups G with $|Z(G)|$ even, $29 \mid |G|$ and a faithful irreducible character of degree $d \le 112$ in characteristic 0.

G	degrees $d \le 112$ (number of characters)	source
$\mathrm{SL}_2(29)$	14 (2), 28 (7), 30 (7)	[HM01, Table 2(b)]
$\mathrm{SL}_2(59)$	30 (2), 58 (15), 60 (14)	[HM01, Table 2(c)]
$\mathrm{SL}_2(173)$	86 (2)	[HM01, Table 2(b)]
2. Ru	28 (2)	[HM01, Table 3], row $d = 28$

The list can also be checked without Table 3. By [HM01, §2.1] (the bounds of Landazuri–Seitz–Zalesskii), every simple group of Lie type whose covering groups have a nontrivial representation of degree ≤ 250 in non-defining characteristic is in an explicit finite list. In that list only $L_2(q)$ and $S_4(17)$ have order divisible by 29 (GAP), and the representations of $S_4(17)$ and $2.S_4(17) = \mathrm{Sp}_4(17)$ in Table 3 have degree

≥ 144 (they include the Weil representations). Among the sporadic groups, 29 divides the order only of Ru, J_4, Fi'_{24} and M, and Fi'_{24} and M do not occur in the tables at all [HM01, §4].

Exclusion of the other groups. $\mathrm{SL}_2(59)$ and $\mathrm{SL}_2(173)$ do not contain S (G-F5). Suppose $\bar{Q} = 2$. Ru. The image of S in Ru is isomorphic to $\mathrm{PSL}_2(29)$; it lies in a maximal subgroup of order divisible by 29, which is $L_2(29)$ (G-F4), and so equals it. Hence S is the full preimage of this $L_2(29)$ in 2. Ru (both have order 24 360). In particular this preimage is $\mathrm{SL}_2(29)$ and not $2 \times L_2(29)$; GAP also finds no class fusion of $2 \times L_2(29)$ into 2. Ru (G-F12 in `out_facts.g` [C25]). Then $X|_S$ is irreducible of degree 28 (G-F4), not a sum of characters of degree 14, a contradiction. Hence $\bar{Q} \cong \mathrm{SL}_2(29)$. As S embeds in $\bar{Q}$ and both have order 24 360, $Q_1 = S \ker X$ with $\ker X$ central, and $Q_1 = [Q_1, Q_1] = S$.

Thus $E(A) = S$, which is characteristic in A; so $S \trianglelefteq A$, $A = N_A(S)$ and $|A| = 1\,169\,280$ by Theorem B(b). □

What the proof relies on. (1) Finite computations: G-F1–G-F5, Theorem B(a), (b), and the diagonal check of F.4(c) [C24, C25]. G-F4 uses the character tables of Ru, 2. Ru, $2.L_2(29)$ in GAP's library CTblLib; the others use only the groups and the lattice. (2) The classification of finite simple groups, through [HM01, Theorem 1] and Table F1, and through the Schreier conjecture in F.4(b). The list of simple groups in the rows of degree $\leq$ 112 of [HM01, Table 3] and [HM02, Table 2], with a check of their orders, is `code/supplementary/automorphisms/out_facts.g`. (3) Standard facts of finite group theory: Bender, Thompson's critical subgroup, Burnside, Clifford, Roquette [Asc00]. To reproduce (1), run `bash code/run_supplementary.sh aut_normalizer aut_diagonal group_facts out_facts` in the SageMath environment of `Dockerfile.sage` or `environment.yml` (with GAP and CTblLib); this takes about 15 minutes (12 minutes measured). Each step compares its results with the values stated here and ends with a nonzero exit code if any differs.

Appendix G. Comparison with known constructions

Table G1. Selected constructions and counting results for extremal even unimodular lattices, following [Sak26, §1], the original papers and the Catalogue of Lattices [NS] (as of October 2026).

dimension n	$\mu(n)$	examples or counting results	sources
8	2	E_8 (unique)	classical
16	2	$E_8 \perp E_8$, D_{16}^+ (all)	classical
24	4	Leech lattice (unique)	[SPLAG]
32	4	more than 10^7	mass formula by root system [Kin03, Cor. 5]
40	4	more than 10^{51}	mass formula and Siegel's theorem [SSP99]; [Sak26, §1]
48	6	at least 4: P_{48p}, P_{48q}, P_{48n}, P_{48m}	Leech–Sloane 1971 [LS71]; Nebe [Neb98, Thm. 5.3], [Neb14]
56	6	several	Ozeki 1989 [Oze89]; Batut–Quebbemann–Scharlau [BQS95]; Nebe [Neb98]
64	6	several	Quebbemann [Que84]; Nebe [Neb98, §5]; Shimada [Shi18, Shi22]
72	8	at least 1 (Γ_{72})	Nebe 2010 [Neb12]; the only one from Griess's construction [NP14]
80	8	at least 4	Bachoc–Nebe 1998 [BN98]; [SW10]; [Wat12]
88	8	at least 6	Sakamoto 2026 [Sak26] (2), Gauvin 2026 [Gau26] (4), both preprints; [NS]
96	10	none known	open
104	10	none known	open
112	**10**	$\boldsymbol{L_{112}}$ **(this paper)**	this paper
120	12	none known	open